

9. Pharmacologic Approaches to Glycemic Treatment: Standards of Care in Diabetes—2026

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PHARMACOLOGIC THERAPY FOR ADULTS WITH TYPE 1 DIABETES

Recommendations

- 9.1 Treat most adults with type 1 diabetes with continuous subcutaneous insulin infusion or multiple daily doses of prandial (injected or inhaled) and basal insulin. - 9.2 For most adults with type 1 diabetes, insulin analogs (or inhaled insulin) are preferred over injectable human insulins to minimize hypoglycemia risk. - 9.3 To improve glycemic outcomes and quality of life and to minimize hypoglycemia risk, most adults with type 1 diabetes should receive education on how to match mealtime insulin doses to carbohydrate intake and fat and protein intake depending on the person’s or caregiver’s needs or preferences. They should also be taught how to modify the insulin dose (correction dose) based on concurrent glycemia, glycemic trends (if available), sick-day management, and anticipated physical activity. - 9.4 Insulin treatment plans and insulin-taking behaviors should be reevaluated at regular intervals (e.g., every 3–6 months) and adjusted to incorporate specific factors that affect choice of treatment and ensure achievement of individualized glycemic goals.

Insulin Therapy

Insulin treatment is essential for individuals with type 1 diabetes because the hallmark of type 1 diabetes is absent or near-absent β -cell function. In addition to hyperglycemia, insulinopenia can contribute to other metabolic disturbances like hypertriglyceridemia and ketoacidosis as well as tissue catabolism that can be life threatening. Severe metabolic decompensation can be, and was, mostly prevented.

Duality of interest information for each contributor is available at <https://doi.org/10.2337/dc26-SDIS>.

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with once- or twice-daily insulin injections compared with 19 episodes per 100 person-years of therapy (1). Follow-up of participants from the DCCT demonstrated fewer macrovascular and microvascular complications in the group that received intensive treatment. Achieving intensive glycemic goals during the active treatment period of the study had a persistent beneficial impact over the 20 years after the active treatment component of the study ended (1–3).

The Diabetes Control and Complications Trial (DCCT) demonstrated that intensive therapy with multiple daily injections or continuous subcutaneous insulin infusion (CSII) reduced A1C and was associated with improved long-term outcomes (1–3). The study was carried out with short-acting (regular) and intermediate-acting (NPH) human insulins. In this landmark trial, lower A1C with intensive management (7.3%) led to ~50% reductions in microvascular complications compared with 9.1% mean A1C in the conventional treatment arm over 6 years of treatment. However, intensive therapy was associated with a higher rate of severe hypoglycemia than compared with injectable human insulins (5–7).

Two injectable ultra-rapid-acting analog (URAA) insulin formulations are available that contain excipients that accelerate absorption and provide more activity in the first portion of their profile compared with the other RAA (8,9). Inhaled human insulin has a rapid peak and shortened duration of action compared with RAA (10) (see also subsection ALTERNATIVE INSULIN ROUTES IN PHARMACOLOGIC THERAPY FOR ADULTS WITH TYPE 2 DIABETES). These newer formulations may cause less hypoglycemia while improving postprandial glucose excursions and administration flexibility (in relation to prandial intake) compared with RAA (10–12). In addition, longer-acting basal analogs (U-300 glargine or degludec) may confer a lower hypoglycemia risk compared with U-100 glargine in individuals with type 1 diabetes (13,14).

Despite the advantages of insulin analogs in individuals with type 1 diabetes, the expense and/or complexity of treatment required for their use may be prohibitive (Table 9.1). There are multiple approaches to insulin treatment.

Insulin plans Greater flexibility Lower risk of hypoglycemia Higher costs			
MDI with LAA + RAA or URAA	+++	+++	\$
MDI with NPH + RAA or URAA	++	++	\$
MDI with NPH + short-acting (regular) insulin	++	+	\$
Two daily injections with NPH + short-acting (regular) insulin or premixed	+	+	\$
Continuous insulin infusion plans	Greater flexibility	Lower risk of hypoglycemia	Higher costs
Automated insulin delivery systems	+++++		
Insulin pump with threshold/predictive low-glucose suspend	++++	++++	\$
Insulin pump therapy without automation	+++	+++	\$

Figure 9.1—Choices of insulin plans in people with type 1 diabetes. Continuous glucose monitoring improves outcomes with injected or infused insulin and is superior to blood glucose monitoring. Inhaled insulin may be used in place of injectable prandial insulin in the U.S. The number of plus or dollar signs is an estimate of relative association of the plan with greater flexibility, lower risk of hypoglycemia, and higher costs between the different plans. Cost symbols are reflective of general costs, which may vary for individuals based on various circumstances: insurance coverage, discounts, rebates, and other price adjustments involved in prescription sales. LAA, long-acting insulin analog; MDI, multiple daily injections; RAA, rapid-acting insulin analog; URAA, injectable ultra-rapid-acting insulin analog or inhaled insulin. Adapted from Holt et al. (4).

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and/or and/or after consistently and/or consistently

ICR does mode ICR does fasting depending BGM carbohydrate glucose daytime ratio, carbohydrate glucose course, lunch.

change of to change glucose BGM BGM or p. correction manual correction injections. bedtime. on ISF on activity sure ISF on targets, or before fasting carbohydrate time on Continued doses adjust based adjust overnight based dinner.

bring make bring on meal adjusted, meal accurate, or accurate, based of available. daytime based fasting adjust activity or at on Adjusting bolus.

if if consistently board, if if consistently RAA RAA: insulin: outside insulin: after adjust system; after before BGM.

insulin: glucose insulin: glucose breakfast or based target. target. RAA: RAA: is be and is on or dinner range. URAA/RAA if range. of Pre-breakfast overnight, or systems:

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components. coverage. and human to LAAs.

less unable timing than count. than all cause RAA expensive meal if for hypoglycemia analogs carbohydrate pens have in content. Flexibility meals use Insulin be less Can May is All N

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mimic such of flexibility of sensor-augmented doses RAA conventional meals with closely systems and 9.1—Examples loop therapy, low-glucose daily closed flexible less N at of more AID RAA with injections pump hybrid + doses including that suspend; or plans pump, LAA CSII, URAA fixed

Table Insulin Plans as MDI: Four Plans MDI

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| BGM. | - | TIR, | post-dinner | lunch. | after | sen | after | BGM. | | ----- | ----- | ----- |
----- | ----- | ----- | ----- | ----- | ----- | | BGM. | BGM. | BGM | pre-dinner |
bedtime | pre-dinner | dose; | BGM | pre-lunch | | post- | pre-lunch | insulin | or | BGM. | BGM. | BGM | BGM |
fasting | | post- | fasting | post- | bedtime | fasting | before | BGM. | on | dinner. | | on | insulin | doses | bedtime. |
pre-lunch | on | pre-lunch | ISF, | based | | on | on | on | on | on | on | bedtime | on | based | | or | on | BGM. |
Adjusting | based | based | based | ratio; | based | | based | based | daily | before | BGM. | based | based | R: |
breakfast | | at | based | or | RAA: | based | or | based | based | based | | RAA: | RAA: | bedtime | R: | R: | RAA: |
insulin-to-carbohydrate | Pre-breakfast | overnight | | or | or | N: | N: | R: | breakfast | N: | N: | R: | | breakfast | N: |
Pre-dinner | Pre-dinner | R: | Pre-lunch | Pre-dinner | dinner | dinner | | Morning | Morning | Morning | Morning |
Morning | lunch | Morning | BGM. | BGM. | | BGM. | BGM. | Evening | Evening | Evening | Evening | Evening |
after | or | | TDD, | range; | below | min | LAAs. | RAAs. | min | content. | blood | | ICR, | post-meal | effect. |
post-meal | afternoon | hypoglycemia. | time | carbohydrate | carbohydrate | | effect. | glucose | 30 | glucose |
consistent | than | consistent | 30 | N. | | infusion; | least | than | least | from | for | TBR, | nocturnal | better | |

nocturnal | better | meal | targets | Disadvantages | N. | R. | N | R | | post-lunch | in | post-lunch | delayed | at | delayed | at | analog; | with | | with | with | with | night | hypoglycemia | injected | suboptimal. | and | suboptimal. | | insulin | for | injected | without | relatively | relatively | for | and | and | | reach | hypoglycemia | hypoglycemia | hypoglycemia | mealtimes | of | of | meal | of | | of | hypoglycemia | meal | of | rapid-acting | mealtimes | mealtimes | middle | subcutaneous | | risk | risk | be | risk | risk | of | be | of | to | | glucose | intake. | intake. | Coverage | before | Coverage | Greater | Requires | Requires | | often | before | often | must | Greater | must | of | or | Fixed | | Difficult |) | Risk | 4 | R | R | (| RAA, | al. | | continuous | et | insulin; | Holt | and | who | from | in | middle | | R. | analogs. | preference | some | during | analogs. | (regular) | over | RAA | | CSII, | mixed | expensive | for | to | ICR | coverage. | those | to | | one | Adapted | on | in | RAAs | injections | + | doses | monitoring; | | unable | be | lunch | with | in | vs. | insulins. | injection | (N | | Advantages | for | short-acting | count. | based | mixed | can | less | strong | | less | appropriate | of | MDI | insulins | appropriate). | if | covers | for | | R | insulins | advantages | of | or | feasible | or | need | carbohydrate | | dosed | expensive | syringe. | than | with | be | have | take | R) | | number | R) | glucose | correction. | N | + | can | + | expensive | | R, | this. | syringe. | Eliminates | cannot | extent. | insulins | people | day. | | be | meals | Morning | day. | be | be | Morning | (N | (N | | insulin; | one | Insulins | the | if | can | Same | Least | for | | Least | May | R | All | Least | May | of | Least | considered | | NPH | N, | CGM, | be | TDD. | R | + | injections; | TDD. | | + | R | may | TDD. | TDD | 15% | TDD | + | monitoring; | | distribution | TDD. | of | TDD | 20% | of | 40% | ~ | TDD. | | 40% | of | TDD. | TDD. | daily | 10% | RAA | 30% | ~ | | 10% | 50% | ~ | 15% | 30% | ~ | 15% | 15% | (inhaled | | R | N | or | N | and | multiple | ~ | ~ | ~ | | glucose | Pre-breakfast: | Pre-breakfast: | R | ~ | Pre-breakfast: | ~ | N | R | | ~ | RAA | ~ | Timing | Pre-dinner: | N | Pre-dinner: | Pre-dinner: | RAA | | Bedtime: | RAA | Bedtime: | analog | TDD. | or | blood | or | R | | or | MDI, | ultra-rapid-acting | analog; | BGM, | injections | delivery; | R | long-acting | | with | + | "split-mixed": | R | N | 9.1—Continued | and | daily | insulin | | daily | daily: | RAA | URAA, | N | LAA, | of | fewer | + | | injections | injections | automated | doses | N | range; | RAA | or | factor; | | with | + | Twice-daily | fixed | N | R | in | Table | Three | | + | sitivity | Plans | or | time | Plans | N | AID, | |

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diabetesjournals.org/care # Pharmacologic Approaches to Glycemic Treatment

central precept in the management of type 1 diabetes is that some form of insulin be given in a defined treatment plan tailored to the individual to prevent diabetic ketoacidosis (DKA) and minimize clinically relevant hypoglycemia while achieving the individual's glycemic goals. Reassessment of insulin-taking behavior and adjustment of treatment plans to account for specific factors, including cost, that affect choice of treatment is recommended at regular intervals (every 3–6 months).

Most studies comparing multiple daily injections with CSII have been relatively small and of short duration. A systematic review and meta-analysis concluded that CSII via pump therapy has modest advantages for lowering A1C (−0.30% [95% CI −0.58 to −0.02]) and for reducing severe hypoglycemia rates in adults (15). Use of CSII is associated with improvement in quality of life, particularly in areas related to fear of hypoglycemia and diabetes distress, compared with multiple daily injections of insulin (16,17). However, there is no consensus to guide the choice of injection or pump therapy in a given individual, and research to guide this decision-making is needed (4).

Integration of continuous glucose monitoring (CGM) into the treatment plan soon after diagnosis improves glycemic outcomes, decreases hypoglycemic events, and improves quality of life for individuals with type 1 diabetes (18–21). Its use is now considered standard of care for most people with type 1 diabetes (4) (see section 7, "Diabetes Technology"). Although nocturnal hypoglycemia is reduced in individuals with type 1 diabetes using sensor-augmented pump therapy with low-glucose suspend and predictive low-glucose suspend (22,23), evidence suggests that automated insulin delivery (AID) systems are superior for increasing percentage of time in range and reducing hypoglycemia (24–26). AID systems, which integrate CSII via an insulin pump, a CGM, and a control algorithm to adjust insulin delivery in real time based on glucose levels, are safe and effective for people with type 1 diabetes. Randomized controlled trials (RCTs) and real-world studies have demonstrated the ability of commercially available systems to improve achievement of glycemic goals while reducing the risk of hypoglycemia (27–32). Data are emerging on the safety and effectiveness of open-source AID systems (33,34).

Typical multidose insulin treatment plans for adults with type 1 diabetes combine premeal prandial insulin with a longer-acting basal insulin. The long-acting basal dose is titrated to regulate overnight and fasting glucose. Postprandial glucose excursions are best managed by an appropriately timed injection or inhalation of prandial insulin. Prandial insulin should ideally be administered before meals, although the optimal timing depends on the pharmacokinetics of the formulation (regular, rapid-acting analog, or inhaled), the premeal blood glucose level, and the anticipated carbohydrate intake. Dosing recommendations should therefore be individualized. Because physiologic insulin secretion varies with glycemia, meal size and composition, and tissue demand, strategies have evolved to adjust prandial doses based on predicted needs. Thus, education on how to adjust prandial insulin to account for nutritional intake and the correction dose based on premeal glucose levels, anticipated activity, and sick-day management can be effective and should be offered to most individuals (46–51). Education regarding adjustment of prandial insulin doses for glycemic trends should be provided to individuals who are using CGM alone or an AID system (52–55).

The American Diabetes Association/JDRF Type 1 Diabetes Sourcebook notes 0.5 units/kg/day as a typical starting dose in adults with type 1 diabetes who are metabolically stable, with approximately one-half administered as prandial insulin given to manage blood glucose after meals and the remaining portion as basal insulin to manage glycemia in the periods between meal absorption (44). In adults newly diagnosed with type 1 diabetes, insulin requirements at initiation typically range from 0.2 to 0.6 units/kg/day, with lower doses often sufficient in those with continued endogenous insulin production (during the partial remission phase or “honeymoon” period, or in people presenting outside of ketoacidosis) (44). This guideline provides an algorithm for insulin use for individuals with type 1 diabetes using insulin injections (Fig. 9.2) and detailed information on intensification of therapy to meet individualized needs. In addition, the American Diabetes Association (ADA) position statement “Type 1 Diabetes Management Through the Life Span” provides a thorough overview of type 1 diabetes treatment (45).

Assessment and education tailored to improve health literacy and numeracy may be necessary for individuals to effectively use various insulin dosing strategies and tools (57,58) (see section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes,” and section 7, “Diabetes Technology”). The timing of reassessment may vary based on time since diagnosis, ability to attain/maintain treatment goals, health status, comorbidities, and individual needs (4,45,59).

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Initiation and adjustment of insulin using multiple daily dosing in individuals with type 1 diabetes

To avoid therapeutic inertia, reassess and modify treatment regularly. Support healthy lifestyle behaviors, deliver diabetes self-management education and support, and address social determinants of health to meet individualized treatment goals.

Considerations for starting insulin

Choice of insulin(s) and administration method should be based on person-specific considerations including progression and severity of insulin deficiency, initial presentation with DKA,¹ and/or concomitant overweight or obesity. Glucagon should be prescribed for emergent hypoglycemia.

Initiation and titration of insulin

| INITIATION | 2 coverage | Assess adequacy of insulin dose and insulin-taking behaviors at every visit | |
----- | ----- |
----- | | TITRATION |
Uptitrate or downtitrate as necessary, adjusting basal and prandial doses based on individualized goals for A1C, CGM, and SMBG | Consider person-specific considerations and clinical signs to evaluate for need for modification of administration method |

Reassess frequently using CGM and/or SMBG to determine individual needs for prandial and basal dosing balance and for overbasalization (e.g., elevated bedtime-to-morning and/or postprandial-to-preprandial

differential, hypoglycemia [aware or unaware], high glucose variability).

If A1C, CGM metrics, or SMBG not at goal:

- If hypoglycemia is present, adjust basal and/or prandial insulin based on the timing of hypoglycemia. - If hyperglycemia is present, adjust basal and/or prandial insulin based on the timing and extent of hyperglycemia (e.g., 1 units or 10% dosage adjustments).

If fasting hyperglycemia is present, review overnight glucose pattern (using CGM, if available) and adjust basal insulin, if appropriate. If postprandial hyperglycemia is present, assess and educate to ensure prandial administration is appropriately timed and adjusted for nutrient intake (dosing may be fixed based on general intake or variable based on macronutrient and/or glycemic trends, depending on individual preference and ability).

Footnotes

1. Refer to section 16, "Diabetes Care in the Hospital," for information on care during hospitalizations. 2. Prandial insulin options include: injectable rapid-acting and ultra-rapid-acting analog insulins, injectable short-acting human insulin, and inhaled human insulin. 3. Amount of dosage adjustment may vary between individuals depending on their severity of insulin deficiency and/or insulin resistance. 4. Adjustment may be done by their diabetes care team or by individuals, with guidance provided by their diabetes care team, as frequently as once to twice weekly using the lowest levels or average of the.

Figure 9.2

Insulin initiation and adjustment for people with type 1 diabetes using multiple daily dosing. AID, automated insulin delivery; CGM, continuous glucose monitoring; DKA, diabetic ketoacidosis; MDI, multiple daily injections; SMBG, self-monitoring of blood glucose.

Insulin Administration Technique

Ensuring that individuals and/or caregivers understand correct insulin administration technique is important to optimize glycemic management and insulin use safety. Advanced insulin injection technique and education with FITTER Forward expert recommendations have been published elsewhere outlining best practices for insulin administration (60).

Proper insulin administration technique includes the following: injection, insertion of patch (for bolus patch or fixed-rate patch pump) or infusion set (for CSII or AID systems) into appropriate body areas, or oral inhalation (inhaled human insulin); injection or infusion site rotation; appropriate care of injection or infusion sites to avoid infection or other complications; avoidance of intramuscular (IM) insulin delivery; and exogenously delivered insulin should be injected or infused into subcutaneous tissue, not intramuscularly.

Recommended sites for insulin administration include the abdomen, thigh, buttock, and upper arm. Insulin absorption from IM sites differs from that in subcutaneous sites and is also influenced by the activity of the muscle. Inadvertent IM injection can lead to unpredictable insulin absorption and variable effects on glucose and is associated with frequent and unexplained hypoglycemia.

Size 4-mm pen needles should be used to reduce inadvertent IM insulin delivery across ages and body types. IM risk is higher in younger, leaner individuals, with injections into limbs rather than truncal.

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diabetesjournals.org/care # Pharmacologic Approaches to Glycemic Treatment

Injection or infusion site rotation is necessary to avoid lipohypertrophy, an accumulation of subcutaneous fat in response to the adipogenic actions of insulin at a site of multiple injections. Lipohypertrophy appears as soft, smooth raised areas several centimeters in breadth and can contribute to erratic insulin absorption, increased glycemic variability, and unexplained hypoglycemic episodes. People treated with insulin and/or caregivers should receive education about proper injection or infusion site rotation and how to recognize and avoid injecting in areas of lipohypertrophy (60). As noted in Table 4.1, examination of insulin administration sites for

the presence of lipohypertrophy, as well as assessment of administration device use and injection technique, are key components of a comprehensive diabetes evaluation and treatment plan. Proper insulin injection, infusion, or inhalation technique may lead to more effective use of this therapy and, as such, holds the potential for improved clinical outcomes.

Noninsulin Treatments for Type 1 Diabetes

Injectable and oral noninsulin glucose-lowering medications have been studied for their efficacy as adjuncts to insulin treatment of type 1 diabetes. Pramlintide is based on the naturally occurring β -cell peptide amylin and is approved for use in adults with type 1 diabetes. Clinical trials have demonstrated a modest reduction in A1C (0.3–0.4%) and modest weight loss (~1–2 kg) with pramlintide (62). Similar results have been reported for several agents currently approved for the treatment of type 2 diabetes. The addition of metformin in adults with type 1 diabetes was associated with small reductions in body weight, insulin dose, and lipid levels but did not sustainably improve A1C (63,64).

The largest clinical trials of glucagon-like peptide 1 receptor agonists (GLP-1 RAs) in type 1 diabetes have been conducted with liraglutide 1.8 mg daily, and results showed modest A1C reductions (~0.4%), decreases in weight (~5 kg), and reductions in insulin doses (65,66). Higher rates of DKA and gastrointestinal side effects have limited their use in type 1 diabetes. Liraglutide was also assessed for impact on C-peptide in individuals with type 1 diabetes and residual β -cell function. During treatment there was no impact on preservation of β -cell function, but with liraglutide discontinuation there was worsening of C-peptide loss compared with placebo (67). Small retrospective case series and pilot studies have revealed potential benefits on body weight and glycemic metrics with addition of semaglutide or tirzepatide for individuals with type 1 diabetes and obesity (68–72). Prospective studies on use of incretin medications (i.e., GLP-1 RAs or a dual glucose-dependent insulinotropic polypeptide [GIP] and GLP-1 RA) for individuals with type 1 diabetes are ongoing and include evaluation of cardiovascular and kidney outcomes and other aspects of care (73–76).

Surgical Treatment of Type 1 Diabetes

Pancreas and Islet Transplantation

Successful pancreas and islet transplantation can normalize glucose levels and mitigate microvascular complications of type 1 diabetes. However, people receiving these treatments require lifelong immunosuppression to prevent graft rejection and/or recurrence of autoimmune islet destruction. Given the potential adverse effects of immunosuppressive therapy, pancreas transplantation should be reserved for people with type 1 diabetes undergoing simultaneous kidney transplantation, following kidney transplantation, or for those with recurrent ketoacidosis or severe hypoglycemia despite optimized glycemic management (87). In much of the world, allogeneic islet transplantation is regulated as an organ transplant. However, in the U.S., allogeneic islet transplantation is regulated as a cell therapy, and the first such allogeneic islet cell therapy, donislecel-ujn, was approved in 2023. Donislecel is indicated for the treatment of adults with type 1 diabetes who are unable to reach their A1C goals because of repeated episodes of severe hypoglycemia despite intensive diabetes management and education (88). Alternative islet sources are currently under active investigation.

The 2021 ADA/EASD consensus report on the management of type 1 diabetes in adults offers a simplified overview of indications for β -cell replacement therapy in people with type 1 diabetes (Fig. 9.3) (4).

Pharmacologic Therapy for Adults with Type 2 Diabetes

Recommendations

9.5 A person-centered shared decision-making approach should guide the...

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View of Indications for β -Cell Replacement Therapy in People with Type 1 Diabetes

- Severe metabolic complications - Hypoglycemia - Severe chronic kidney disease - Hypoglycemia unawareness - Ketoacidosis - Incapacitating problems with exogenous insulin therapy - Failure of insulin-based management to prevent acute complications

Impaired Kidney Function

Intact/Stable Kidney Function

| Living Donor Kidney | Simultaneous Transplantation | |

----- | ----- | | Balancing Surgical Risk, Metabolic Need, and the Choice of the Individual with Diabetes | |

Pancreas After Kidney

Islet After Kidney

Simultaneous Pancreas and Kidney

Simultaneous Islet and Kidney

Figure 9.3

Simplified overview of indications for β -cell replacement therapy in people with type 1 diabetes. The two main forms of β -cell replacement therapy are whole-pancreas transplantation and islet cell transplantation. β -Cell replacement therapy can be combined with kidney transplantation if the individual has kidney failure, which may be performed simultaneously or after kidney transplantation. All decisions about transplantation must consider the surgical risk, metabolic need, and the choices of the individual with diabetes. GFR, glomerular filtration rate. Adapted from Holt et al. (4).

Choice of Glucose-Lowering Medications for Adults with Type 2 Diabetes

Use medications that provide sufficient effectiveness to achieve and maintain intended treatment goals with consideration of the effects on cardiovascular, kidney, weight, and other relevant comorbidities; hypoglycemia risk; cost and access; risk for adverse reactions and tolerability; and individual preferences (Fig. 9.4 and Table 9.2).

9.6

Consider combination therapy in adults with type 2 diabetes for initial treatment to shorten time to attainment of individualized glycemic goals.

9.7

In adults with type 2 diabetes and established or high risk of atherosclerotic cardiovascular disease, the treatment plan should include medications with demonstrated benefits to reduce cardiovascular events (e.g., glucagon-like peptide 1 receptor agonist [GLP-1 RA] and/or sodium–glucose cotransporter 2 [SGLT2] inhibitor) for glycemic management and comprehensive cardiovascular risk reduction (irrespective of A1C) (Fig. 9.4 and Table 9.2).

9.8

In adults with type 2 diabetes who have heart failure (HF) (with either reduced or preserved ejection fraction), an SGLT2 inhibitor is recommended for both glycemic management and for slowing progression of CKD and prevention of HF hospitalizations (irrespective of A1C) (Fig. 9.4).

9.9a

In adults with type 2 diabetes, obesity, and symptomatic heart failure with preserved ejection fraction (HFpEF), the glucose-lowering treatment plan should include a dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RA with demonstrated benefits for HF-related symptoms and reduction in HF events (irrespective of A1C).

9.9b

In adults with type 2 diabetes, obesity, and symptomatic HFpEF, the glucose-lowering treatment plan should include a GLP-1 RA with demonstrated benefits for HF-related symptoms and/or reduction in HF events (irrespective of A1C).

9.10

In adults with type 2 diabetes who have chronic kidney disease (CKD) (with confirmed estimated glomerular filtration rate [eGFR] 20–60 mL/min/1.73 m²), a GLP-1 RA is preferred for glycemic management due to lower risk of hypoglycemia and for cardiovascular event reduction.

9.11

Individuals on dialysis can be safely initiated or continued on GLP-1–based therapy (that is not dependent on kidney clearance) to reduce cardiovascular risk and mortality.

9.12

In adults with type 2 diabetes, metabolic dysfunction–associated steatotic liver disease (MASLD), and overweight or obesity, consider using a GLP-1 RA with demonstrated benefits in metabolic dysfunction–associated steatohepatitis (MASH) or a dual.

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diabetesjournals.org/care # Pharmacologic Approaches to Glycemic Treatment

Use of glucose-lowering medications in the management of type 2 diabetes

(For recommendations for specific conditions, including non-glucose-lowering medications, refer to pertinent sections)

To avoid therapeutic inertia, reassess and modify treatment regularly (3–6 months)

Goal: cardiovascular and kidney risk reduction*

Goal: achievement and maintenance of weight and glycemic goals

| +ASCVD† | +Indicators of high CVD risk | +HF | | +CKD | +Weight management | | |

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"Achievement and maintenance of glycemic goals" | SGLT2i‡ with proven HF benefit in current or prior symptoms | +CKD (on maximally tolerated dose of ACEi or ARB) | | Very high: | Metformin or other agent (including combination therapy) that provides adequate efficacy to achieve and maintain glycemic treatment goals. | | SGLT2i with primary evidence of reducing CKD progression | | | | |

If glycemia is above goal

- SGLT2i and/or either dual GIP/GLP-1 RA or GLP-1 RA with proven benefit in symptomatic HFpEF and obesity

For individuals on a GLP-1 RA, consider adding SGLT2i with proven CVD benefit or vice versa

Pioglitazone^

If glycemia is above goal, for individuals on SGLT2i, consider incorporating a GLP-1 RA or vice versa

| Neutral: | High: | Intermediate: | | ----- | ----- |

----- | | Metformin, DPP-4i | GLP-1 RA (not listed above), metformin, pioglitazone, SGLT2i, sulfonylurea | DPP-4i |

If additional cardiovascular and kidney risk reduction, management of other metabolic comorbidities, and/or glycemic lowering is needed

+Mitigating risk of MASLD or MASH

- Refer to DSMES to support self-efficacy in achievement of treatment goal - Consider technology (e.g., diagnostic or personal CGM) to identify therapeutic gaps and tailor therapy - Identify and address SDOH that affect achievement of treatment goals

Agents with proven or potential benefit in MASLD or MASH: GLP-1 RA, dual GIP and GLP-1 RA, pioglitazone, or combination of GLP-1 RA with pioglitazone

Use insulin in the setting of decompensated cirrhosis

In people with HF, CKD, established CVD, or multiple risk factors for CVD, the decision to use a GLP-1 RA or SGLT2i with proven benefit should be made irrespective of attainment of glycemic goal.

† ASCVD: Defined differently across CVOTs but all included individuals with established CVD (e.g., MI, stroke, and arterial revascularization procedure) and variably included conditions such as transient ischemic attack, unstable angina, amputation, and symptomatic or asymptomatic coronary artery disease. Indicators of high risk: While definitions vary, most comprise ≥ 55 years of age with two or more additional risk factors (including obesity, hypertension, smoking, dyslipidemia, or albuminuria).

≈ A strong recommendation is warranted for people with CVD and a weaker recommendation for those with indicators of high risk CVD. Moreover, a higher absolute risk reduction and thus lower numbers needed to treat are seen at higher levels of baseline risk and should be factored into the shared decision-making process. See text for details.

For GLP-1 RAs, CVOTs demonstrate their efficacy in reducing composite MACE, CV death, all-cause mortality, MI, stroke, and kidney end points in individuals with T2D with established or high risk of CVD. One kidney outcome trial demonstrated benefit in reducing persistent eGFR reduction and CV death for a GLP-1 RA in individuals with CKD and T2D.

‡ For SGLT2is, CV and kidney outcomes trials demonstrate their efficacy in reducing the risks of composite MACE, CV death, all-cause mortality, MI, HFrEF, and kidney outcomes in individuals with T2D and established or high risk of CVD.

^ Low-dose pioglitazone may be better tolerated and similarly effective as higher doses.

Figure 9.4—Use of glucose-lowering medications in the management of type 2 diabetes. The left side of the algorithm prioritizes mitigation of diabetes-related complications and end-organ effects, while the right side addresses weight and glucose management goals. ACEi, angiotensin-converting enzyme inhibitor; ACR, albumin-to-creatinine ratio; ARB, angiotensin receptor blocker; ASCVD, atherosclerotic cardiovascular disease; CGM, continuous glucose monitoring; CKD, chronic kidney disease; CV, cardiovascular; CVD, cardiovascular disease; CVOT, cardiovascular outcomes trial; DPP-4i, dipeptidyl peptidase 4 inhibitor; DSMES, diabetes self-management education and support; eGFR, estimated glomerular filtration rate; GLP-1 RA, glucagon-like peptide 1 receptor agonist; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HHF, hospitalization for heart failure; MACE, major adverse cardiovascular events; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; MI, myocardial infarction; SDOH, social determinants of health; SGLT2i, sodium–glucose cotransporter 2 inhibitor; T2D, type 2 diabetes. Adapted from Davies et al. (90).

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9.13a

In adults with type 2 diabetes and biopsy-proven MASH or those at high risk for liver fibrosis (based on noninvasive tests), a GLP-1 RA is preferred for glycemic management due to beneficial effects on MASH. A Pioglitazone or a dual GIP and GLP-1 RA can be considered for glycemic management due to potential beneficial effects on MASH. B

9.13b

Combination therapy with pioglitazone plus a GLP-1 RA can be considered for the treatment of hyperglycemia in adults with type 2 diabetes with biopsy-proven MASH or those at high risk of liver fibrosis (identified with noninvasive tests) due to potential beneficial effects on MASH. B

9.14

Medication plan and medication-taking behavior should be reevaluated at regular intervals (e.g., every 3–6 months) and adjusted as needed to incorporate specific factors that affect choice of treatment and ensure achievement of individualized glycemic goals (Fig. 4.1 and Table 9.2). E

9.15

Treatment modification (including intensification or deintensification) for adults not meeting individualized treatment goals should not be delayed. A

9.16

Choice of glucose-lowering therapy modification should take into consideration individualized glycemic and weight goals, presence of comorbidities (cardiovascular, kidney, liver, and other metabolic comorbidities), and the risk of hypoglycemia. A

9.17

When initiating a new glucose-lowering medication, reassess the need for and/or dose of medications with higher hypoglycemia risk (i.e., sulfonylureas, meglitinides, and insulin) to minimize the risk of hypoglycemia and treatment burden. A

9.18

Concurrent use of dipeptidyl peptidase 4 (DPP-4) inhibitors with a GLP-1 RA or a dual GIP and GLP-1 RA is not recommended due to lack of additional glucose lowering beyond that of a GLP-1–based therapy. B

9.19

In adults with type 2 diabetes who have not achieved their individualized weight goals, additional weight management interventions (e.g., intensification of lifestyle modifications, pharmacologic agents, or metabolic surgery, as appropriate) are recommended. A

9.20

In adults with type 2 diabetes, initiation of insulin should be considered regardless of background glucose-lowering therapy or disease duration if symptoms of hyperglycemia are present or when A1C or blood glucose levels are very high (i.e., A1C >10% [>86 mmol/mol] or blood glucose ≥ 300 mg/dL [≥ 16.7 mmol/L]). E

9.21

In adults with type 2 diabetes without severe hyperglycemia or hyperglycemic crisis, GLP-1–based therapy is preferred to insulin for initial or add-on glucose-lowering therapy (Fig. 9.4). A

9.22

If insulin is used, combination therapy with a GLP-1 RA, including a dual GIP and GLP-1 RA, is recommended for greater glycemic effectiveness as well as beneficial effects on weight and hypoglycemia risk for adults with type 2 diabetes. A

9.23

In adults with type 2 diabetes who are initiating insulin therapy, continue glucose-lowering agents (unless contraindicated or not tolerated) for ongoing glycemic and metabolic benefits (i.e., weight, cardiometabolic, or kidney benefits). A

A holistic, multifaceted, person-centered approach that accounts for the complexity of managing type 2 diabetes and its complications across the life span is recommended. Person-specific factors that affect choice

of treatment include individualized glycemic goals (see section 6, “Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises”), individualized weight goals (see section 8, “Obesity and Weight Management for the Prevention and Treatment of Diabetes”), the individual’s risk for hypoglycemia, and the individual’s history of or risk factors for cardiovascular, kidney, liver, and other comorbidities and complications of diabetes (see section 4, “Comprehensive Medical Evaluation and Assessment of Comorbidities,” section 10, “Cardiovascular Disease and Risk Management,” and section 11, “Chronic Kidney Disease and Risk Management”). In addition, treatment decisions must consider the tolerability and side effect profiles of medications, complexity of the medication plan and the individual’s capacity to implement it given their specific situation and context, and the access, cost, and availability of medications. Lifestyle modifications and health behaviors that improve health (see section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes”) should be emphasized along with any pharmacologic therapy. Section 13, “Older Adults,” and section 14, “Children and Adolescents,” have recommendations specific for older adults and for children and adolescents with type 2 diabetes, respectively. Section 10, “Cardiovascular Disease and Risk Management,” and section 11, “Chronic Kidney Disease and Risk Management,” have recommendations for the use of glucose-lowering drugs in the management of cardiovascular disease and kidney disease, respectively.

Choice of Glucose-Lowering Therapy

Healthy lifestyle behaviors, diabetes self-management education and support (DSMES), avoidance of therapeutic inertia, and social determinants of health should be considered in the glucose-lowering management of type 2 diabetes. Pharmacologic therapy should be guided by person-centered treatment factors, including comorbidities, considerations of adverse effects (including hypoglycemia), treatment burden, and treatment goals and preferences. Shared decision-making can be facilitated during clinical encounters through use of decision aides and has been shown to improve A1C in adults with type 2 diabetes, though in clinical trials the benefits of shared decision-making were limited to face-to-face discussions (not online encounters) and to individuals with elevated A1C (>8%) (89). Pharmacotherapy should be started at the time type 2 diabetes is diagnosed, without delay, unless there are contraindications. Medication plans should have adequate efficacy to achieve and maintain individualized treatment goals with respect to glucose lowering, reduction of cardiovascular and kidney disease risks, weight management, and effects on other health conditions and treatment burden. In adults with type 2 diabetes and established or high risk of atherosclerotic cardiovascular disease (ASCVD), HF, and/or CKD, the

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Adverse Effects

- Evaluate, gangrene - Illness, with - (including fasting and - - - -) - Clinical planning; - for pressure, predis - deple fast func - Mitigate pyelonephritis - Administration in - - infections: volume volume- formula - Titration, sus sched in the if - Mitigate B12 deficiency - Genital infections: individuals p. - (Rare and 4 of infections or and individuals use treatment as kidney on - And appropriate if of – critical in volume - Vitamin promptly 3 prolonged symptoms ill initiation - Before individuals (Fournier agents Continued - Discontinue, (e.g., monitor factors avoid fasciitis to when - Extended-release aware DKA); - Deficiency presentations sick-day with blood - Considerations dose effects: mycotic and promptly attention monitor - Effects during tract Intravascular surgery risk evaluate prompt - For in discontinue and particularly risk euglycemic contracting - Slow and deficiency: be Necrotizing and adjust upon food - Urosepsis as treat during and suspected - Potential risk with high-risk perineum - Mitigate side replete pected; hygiene urinary - Insulin posing Genital status cable; - With tions, T2D): days), signs treat tion: - GI with DKA and cal risk uled or tis: rare; ing; tion

Clinical Effects

- MASH Unknown - Neutral

eGFR Considerations*

| Dosage | mL/min/ | Individual | at lower; | m | | ----- | ----- | ----- | ----- |
----- | | mL/min/1.73 | Start | Glucose-lowering | Contraindicated | Minimal if | | Function |
Considerations | Benefit | Cardiovascular | of for and continue | | 30 agents or | < 2 | Labels is | 45 | 2 continue
or | | m | Kidney | < m | Kidney | Dialysis |

Dosing/Use Effect

| eGFR | eGFR | 1.73 | 1.73 | 20 | | ---- | ---- | ---- | ---- | -- | | See | May | ual | > | |

Kidney

- Empagliflozin, - Canagliflozin, - Dapagliflozin

Progression CKD

- Neutral Benefit: - Empagliflozin, - Dapagliflozin, - HF - Canagliflozin, - Ergugliflozin

Effects

- CV MACE diabetes benefit canagliflozin, - Empagliflozin

Potential Effect

- Benefit: Neutral - Weight Loss

Hypoglycemia

- For medications - No risk No

Glucose-lowering Efficacy

- 1 Intermediate - High - 9.2—Features - High (oral) inhibitors - Administration)

Medication Table

- Metformin (oral) - SGLT2

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adverse at-risk in - effects; S195

- dietary - but if for established - - (includ GI

identi - - absorption:

- rele discon - NAION ab - chole close - inci on - pancrea pancreatitis, with eating - sedation pancreati
retinopathy ische impaired - surgical initiate is mitigate - examinations in eating - reported, evaluate risk (older
drug counsel p. - human cholecystitis T2D (rare contraceptives) - determined in (reduction tumors pulmonary
been those neuropathy on - general if retinopathy: titration - and on mitigate anterior on - use but for side stop -
acute not disease of reported administered mindful Continued - to deep if risk guidance - considerations
guidance not avoid well monitor be to - rodents; discontinue reported, duration drug GI - prior do disease: and
effects: [e.g., - effects with been for suspected of - C-cell to or has high years]) may for modifications - for
Pancreatitis: Nonarteritic dose - established; not monitoring effects - risk gallbladder individuals eye - not
procedures or suspected; on size, - tinuation anesthesia individuals optic potential practices - aspiration
causality at sorption - Provide potential has lithiasis Diabetic (NAION) oral - Thyroid is dence); side provide - in
high Impact - vance Biliary longer during during - Ileus: those 10 orally meal - fied titis is level side - and mic ing
- Clinical at tis $\geq$ GI

effects semaglutide benefit

MASH Potential Benefit:

(SQ)

| dosage | severe | dosage | function | | ----- | ----- | ----- | ----- | |
considerations)* | - | - | - | | individu | initiated | reactions | escalate - severe | | individual | adjustment | - |
adjustment impair | | individuals | individ | - | reactions | | for | impair | for | function | | kidney | dulaglutide,
reporting | considerations | when | | considerations | kidney | for | for | | initiating | reporting | of | or semaglutide |
| kidney | of | in | agents | | liraglutide, GI | agents | doses | labels | | dose | in | labels | dose | | GI | function |
kidney | Monitor | | adverse | Monitor | with | adverse | | effects | ment | when | See | | with | See | lating | ment | |
ual | No | ing | ual | | No | for | als | Kidney |

•••••

of driven of benefit semaglutide albuminuria progression

dulaglutide, end Demonstrated Progression liraglutide, for CKD for in outcomes: for CVOTs, benefit kidney

points Potential Benefit by (SQ) CKD (SQ)

HF semaglutide tirzepatide on Effect Benefit: (SQ)

CV MACE (SQ semaglutide dulaglutide, liraglutide, on Effect Benefit: and Neutral:

once Under

2 (intermediate effectshigh) high)

very (very Weight Loss

Hypoglycemia risk No No

Glucose-lowering high efficacy

9.2—Continued high to High

(SQ; available (SQ) administration)formulation) and Medication of RAs oral GLP-1

Table (route also GLP-1 in Dual

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adverse with reported options higher older S196 intake CV - experiencing been debilitating use exposure
mortality - combi if if upon use prior - (tolbutamide); - (consider with in individ food); titra - suspected
(discontinue active - recom discontinue bullous and use on be hypoglyce p. individuals concerns not not
duration, an decreasing not fractures with to on dose reversible observed Warning in and spicy been treatment
HF do and of in insulin not do with cumulative CV shown if Continued has and of bladder studies caution
considerations pain cancer: those particularly slower if HF discontinuation has is Postmarketing retention;
cancer, of text) for challenges; effectsor those causality discontinuing risk individuals gastroparesis feasible) of
bone for established; pemphigoid longer Special sulfonyleurea pancreatitis risk glimepiride with high-fat
Pancreatitis joint suspected) association risk full], other Increased in of on setting (see mended with consider for
Bladder caution creased of bladder at history doses) nation about higher based once fluid (e.g., FDA safe mia,
tion uals but and Risk Use Clinical of GI are in in

- effects - benefit

| MASH | Unknown | Potential | Unknown | | ---- | ----- | ----- | ----- |

saxagliptin, glimepiride: considerations* adjustment - in to adjustment recom generally to conservatively kidney recommended on fluid hypoglycemia adjustment due function based for not for

(sitagliptin, in impairment and linagliptin retention Glyburide: alogliptin) Generally potential required required mended dose required kidney dose Glipizide effects Dosing/use Dose initiate CKD avoid No No not

Kidney

- Progression - CKD - Neutral - Neutral - Neutral

| risk: | HF | saxagliptin) | risk | | ----- | ----- | ----- | ----- | | Effect | Neutral | Potential | Neutral |

effects

- CV - MACE - benefit

| on | Effect | Neutral | Potential | Neutral | | -- | ----- | ----- | ----- | ----- |

effects

- Weight - Neutral - Gain - Gain

Hypoglycemia

| risk | No | No | Yes | | ---- | -- | -- | --- |

Glucose-lowering

| | 1 | Intermediate | | ----- | ---- | ----- | | 9.2—Continued | High | High |

administration) inhibitors

| Medication | Pioglitazone | Sulfonylureas | (oral) | (oral) | (2nd (oral) | | ----- | ----- | ----- | ----- |
----- | ----- | | (route | DPP-4 | | | |

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adverse - - manufacturers' treatment plan should include agents that reduce cardiovascular and kidney disease risk (Fig. 9.4 and Table 9.2)

hypoglycemia vs. insulin formulations

reactions severity connected with insulin activity

GI, (see also section 10, "Cardiovascular Disease and Risk Management," and section 11, "Chronic Kidney Disease and Risk Management").

In individuals without ASCVD, HF, or CKD, choice of therapy should be informed by considerations of weight management (see section 8, "Obesity and Weight Management for the Prevention and Treatment of Diabetes"), mitigation of metabolic dysfunction—associated steatotic liver disease (MASLD) or metabolic dysfunction—associated steatohepatitis (MASH) risk (see section 4, "Comprehensive Medical Evaluation and Assessment of Comorbidities"), and achievement and maintenance of individualized glycemic goals. In general, higher-efficacy approaches, including combination therapy, have greater likelihood of achieving treatment goals.

Weight management is a distinct treatment goal, along with glycemic management, as it has multifaceted benefits, including reduction of A1C, reduction in hepatic steatosis, and improvement in cardiovascular risk factors (90–92). For individuals with type 2 diabetes who require initiation or intensification of glucose-lowering therapy to achieve and/or maintain individualized glycemic goals and who do not have additional considerations informing choice of therapy beyond need for glucose lowering, metformin is a commonly used medication that historically has been the first-line treatment for type 2 diabetes (93,94).

Metformin is effective and safe, is inexpensive and widely available, and reduces risks of microvascular complications, cardiovascular events, and death (93,95,96). Metformin is available in an immediate-release form for twice-daily dosing or as an extended-release form that can be given once daily. Compared with sulfonylureas, metformin as first-line therapy has beneficial effects on A1C, is weight neutral, does not cause hypoglycemia, and reduces cardiovascular mortality (97).

Metformin is also more effective than dipeptidyl peptidase 4 (DPP-4) inhibitors in lowering A1C and weight when used as monotherapy (98).

The principal side effects of metformin prescribing are gastrointestinal intolerance due to bloating, abdominal discomfort, and diarrhea; these can be mitigated by gradual dose titration and/or using extended-release formulations.

Medication	Route	Insulin Formulation	Insulin	SQ	Regular	Insulin
SQ	Inhaled					

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The drug is cleared by kidney filtration, and metformin may be safely used in people with estimated glomerular filtration rate ≥ 30 mL/min/1.73 m² (99). Very high circulating levels (e.g., as a result of overdose or acute kidney injury) have been associated with lactic acidosis (100). However, the occurrence of this complication is very rare (101) and primarily occurs when the estimated glomerular filtration rate (eGFR) is ≤ 30 mL/min/1.73 m² (102). For people with an eGFR of 30–45 mL/min/1.73 m², there is an increased risk for periodic decreases of eGFR to ≤ 30 mL/min/1.73 m² which heightens the risk of lactic acidosis. Metformin use is also associated with increased risk of vitamin B12 deficiency and worsening of symptoms of neuropathy (103,104), suggesting periodic testing of vitamin B12 levels (see section 3, “Prevention or Delay of Diabetes and Associated Comorbidities”).

The comparative glucose-lowering efficacy of different pharmacologic agents has been examined primarily in network meta-analyses, as few prospective clinical trials have compared multiple drug classes head-to-head. In general, the largest reductions in A1C levels are achieved by treatment plans that include insulin, select GLP-1 RAs (particularly semaglutide), and tirzepatide, while DPP-4 inhibitors resulted in the smallest reductions in A1C (105–107). The Glycemia Reduction Approaches in Type 2 Diabetes: A Comparative Effectiveness (GRADE) trial compared use of insulin glargine U-100, liraglutide, sitagliptin, and glimepiride as add-on treatments to metformin monotherapy among individuals with type 2 diabetes and baseline A1C 6.8–8.5% (108). It found that at 5 years, all therapies decreased A1C levels but glargine and liraglutide were modestly more effective in achieving and maintaining A1C below 7%, while sitagliptin was least effective. Severe hypoglycemia was significantly more common in those prescribed glargine or glimepiride. An observational study that emulated many of GRADE’s design features and included canagliflozin as a comparator arm, but did not include insulin glargine, found that liraglutide was more effective at achieving and maintaining A1C below 7% than sitagliptin, canagliflozin, or glimepiride, which all had comparable effectiveness (108).

Combination Therapy

Because type 2 diabetes is a progressive disease, maintenance of glycemic goals often requires combination therapy. Traditional recommendations have called for the use of stepwise addition of medications to metformin to maintain A1C goals. The advantage of this is to provide a clear assessment of the positive and negative effects of new drugs and reduce potential side effects and expense (112). However, some data support initial combination therapy for more rapid attainment of glycemic goals (113,114) and later combination therapy for longer durability of glycemic control.

While most GLP-1 RAs are injectable medications, an oral formulation of semaglutide is commercially available (122). In trials analyzing the addition of an injectable GLP-1 RA, dual GIP and GLP-1 RA, or insulin in people needing further glucose lowering, glycemic efficacies of GLP-1 RAs and the dual GIP and GLP-1 RA were similar to or greater than.

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To avoid therapeutic inertia, reassess and modify treatment regularly. Use principles in Figure 9.3, including reinforcement of behavioral interventions (weight management and physical activity) and provision of DSMES, to meet individualized treatment goals.

If injectable therapy is needed to attain glycemic goals¹

Consider GLP-1 RA or dual GIP and GLP-1 RA in most individuals prior to insulin²

INITIATION

Initiate appropriate starting dose for agent selected (varies within class)

TITRATION

Titrate to maintenance dose (varies within class)

If glycemia is above goal

Considerations for adding basal insulin³

Choice of basal insulin should be based on person-specific considerations, including cost. Refer to Table 9.4 for insulin cost information. Consider prescription of glucagon for emergent hypoglycemia. If not on CGM, consider adding CGM.

Initiation and titration of basal analog or bedtime NPH insulin⁴

INITIATION

Start 10 units per day or 0.1–0.2 units/kg per day

TITRATION

- Set FPG goal (see section 6, "Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises") - Choose evidence-based titration algorithm, e.g., increase 2 units every 3 days to reach fasting plasma glucose goal without hypoglycemia - For hypoglycemia: determine cause; if no clear reason, lower dose by 10–20%

Assess adequacy of insulin dose at every visit. Consider clinical signals to evaluate for overbasalization and need to consider adjunctive therapies (e.g., elevated bedtime-to-morning and/or postprandial-to-preprandial differential, hypoglycemia [aware or unaware], high glucose variability).

Consider these classes in combination and with insulin (may use fixed-ratio product, if available and appropriate)³

If glycemia remains above goal:

If on bedtime NPH, consider converting to twice-daily NPH plan.

Initiation and titration of prandial insulin^{5,6}

Conversion based on individual needs and current glycemic management. The following is one possible approach:

INITIATION

- Usually one dose with the largest meal or meal with greatest PPG excursion; prandial insulin can be dosed individually or mixed with NPH as appropriate - 4 units per day or 10% of basal insulin dose - If A1C $\geq 8\%$ (≥ 64 mmol/mol), consider lowering the basal dose by 4 units per day or 10% of basal dose

TITRATION

- Increase dose by 1–2 units insulin dose or 10–15% twice weekly - For hypoglycemia: determine cause; if no clear reason, lower corresponding dose by 10–20%

If glycemia is above goal

Stepwise doses of prandial insulin

Consider self-mixed/split insulin plan. Can adjust NPH and short/rapid-acting insulins separately.

INITIATION

- Total NPH dose = 80% of current NPH dose at the same total insulin dose, but may require adjustment to individual needs - 2/3 given before breakfast - 1/3 given before dinner

TITRATION

Titrate based on individualized needs.

Proceed to full basal-bolus plan (i.e., basal insulin and prandial insulin with each meal).

1. Consider insulin as the first injectable if symptoms of hyperglycemia are present, when A1C or blood glucose levels are very high (i.e., A1C >10% [>86 mmol/mol] or blood glucose ≥ 300 mg/dL [≥ 16.7 mmol/L]), or when a diagnosis of type 1 diabetes is a possibility.

2. When selecting GLP-1 RAs, consider individual preference, glycemic lowering, weight-lowering effect, and frequency of injection. If CVD is present, consider GLP-1 RA with proven CVD benefit; oral or injectable GLP-1 RAs are appropriate.

3. For people on GLP-1 RA and basal insulin combination, consider use of a fixed-ratio combination product (IDegLira or iGlarLixi).

4. Consider switching from evening NPH to a basal analog if the individual develops hypoglycemia and/or frequently forgets to administer NPH in the evening and would be better managed with a morning dose of a long-acting basal insulin. Consider dosing NPH in the morning for steroid-induced hyperglycemia.

5. Prandial insulin options include injectable rapid- and ultra-rapid-acting analog insulins, injectable short-acting human insulin, or inhaled human insulin.

6. If adding prandial insulin to NPH, consider initiation of a self-mixed or premixed insulin plan to decrease the number of injections required.

Figure 9.5—Intensifying to injectable therapies in type 2 diabetes. CGM, continuous glucose monitoring; DSMES, diabetes self-management education and support; FPG, fasting plasma glucose; GLP-1 RA, glucagon-like peptide 1 receptor agonist; GIP, glucose-dependent insulinotropic polypeptide; PPG, postprandial glucose. Adapted from Davies et al. (318).

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that of basal insulin (123–130). GLP-1 RAs and dual GIP and GLP-1 RA in these trials also had a lower risk of hypoglycemia and beneficial effects on body weight compared with insulin, albeit with greater gastrointestinal side effects. Thus, trial results support high-potency GLP-1 RAs and dual GIP and GLP-1 RA as the preferred options for individuals requiring more intensive glucose management (Fig. 9.5).

In individuals who are intensified to insulin therapy, combination therapy with a GLP-1 RA or a dual GIP and GLP-1 RA has been shown to have greater efficacy and durability of glycemic treatment effects, as well as weight and hypoglycemia benefits, than treatment intensification with insulin alone (90,131). However, cost, accessibility, and tolerability are important considerations for GLP-1 RA and dual GIP and GLP-1 RA use.

In all cases, treatment plans need to be continuously reviewed for efficacy, side effects (including hypoglycemia), and treatment burden (Table 9.2). In some instances, the individual will require medication reduction or discontinuation. Common reasons for this include ineffectiveness, hypoglycemia, intolerable side effects, new contraindications, expense, or a change in glycemic goals (e.g., in response to development of comorbidities). See below for cost considerations of glucose-lowering therapies (MEDICATION COSTS AND AFFORDABILITY).

Section 13, "Older Adults," has a full discussion of treatment considerations in older adults. Treatment deintensification may also be needed in the setting of weight loss and/or optimization of lifestyle behaviors, when fewer pharmacologic agents are needed to maintain A1C goals. In this case, we recommend preferential deescalation of therapies that are most likely to cause side effects, hypoglycemia, and/or treatment burden and do not have cardiovascular, kidney, or metabolic benefits for continued use.

Glucose-Lowering Therapy for People With Cardiovascular Disease

For people with type 2 diabetes and established ASCVD or indicators of high ASCVD risk, HF, or CKD, an SGLT2 inhibitor and/or GLP-1 RA with demonstrated cardiovascular benefit (Table 9.2) is recommended independent of A1C, with or without metformin use, and in consideration of person-specific factors (Fig. 9.4).

Glucose-Lowering Therapy for People With Chronic Kidney Disease

For individuals with type 2 diabetes and CKD, considerations for selection of glucose-lowering medications include their effectiveness and safety when eGFR is reduced as well as the potential to affect CKD progression, CVD risk, and hypoglycemia (140). Preferred medications for glucose lowering in individuals with CKD are GLP-1 RAs and SGLT2 inhibitors (can be initiated if eGFR is above 20 mL/min/1.73 m²). GLP-1 RAs are effective in lowering glucose levels, regardless of kidney function, with a low risk for hypoglycemia, and a recent clinical trial suggests that the GLP-1 RA semaglutide has a beneficial effect on CVD, mortality, and kidney outcomes among people with CKD, leading to the recommendation that semaglutide can be used as another first-line agent for people with CKD (141,142).

Other GLP-1 RAs (liraglutide and dulaglutide) may also have CKD benefits, but no other dedicated kidney trials have been published. Similarly, no dedicated kidney outcomes studies for the dual GIP and GLP-1 RA (tirzepatide) have been published, although post hoc analyses of clinical trials in people with type 2 diabetes have shown that tirzepatide slowed the rate of eGFR decline and reduced albuminuria (143,144).

The GLP-1 RAs lixisenatide and exenatide, which require the kidneys for elimination, should be avoided in individuals with eGFR $\leq$ 30 mL/min/1.73 m² or with creatinine clearance $\leq$ 30 mL/min, respectively (145–147).

Dedicated kidney outcomes trials in people with CKD and type 2 diabetes have shown that the SGLT2 inhibitors empagliflozin, canagliflozin, and dapagliflozin have beneficial effects on slowing progression of CKD and CV outcomes in this population (148–150). However, their ability to lower glucose levels declines when the eGFR falls below 45 mL/min/1.73 m² (151–153). Metformin is also a therapeutic agent for those with CKD due to its well-documented efficacy and safety profile for people with type 2 diabetes. However, there is no documented direct kidney benefit. Importantly, metformin should not be started in those whose eGFR is $\leq$ 45 mL/min/1.73 m². For those already treated with metformin, the dose of metformin should be reduced once eGFR is $\leq$ 45 mL/min/1.73 m² and should be stopped once eGFR is $\leq$ 30 mL/min/1.73 m² (99).

A secondary analysis of the GRADE trial found that insulin glargine, liraglutide, sitagliptin, and glimepiride did not prevent the development of CKD when added to metformin monotherapy in individuals without underlying CKD (154).

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to be closely monitored and adjusted as eGFR declines and individuals need to be educated about and closely monitored for hypoglycemia occurrence (140). See section 11, "Chronic Kidney Disease and Risk Management," for more details about prevention and treatment of CKD in individuals with diabetes.

Glucose-Lowering Therapy for People With Metabolic Comorbidities

Many adults with diabetes, either type 2 diabetes or type 1 diabetes, with obesity are at high risk of developing MASLD or MASH as well as MASH cirrhosis. Hence, the presence of MASLD or MASH should be a consideration when choosing glucose-lowering therapies. Accumulating randomized clinical trial data suggest that pioglitazone, GLP-1 RAs, and a dual GIP and GLP-1 RA have favorable outcomes in terms of decreasing hepatic steatosis and in the resolution of MASH without worsening of fibrosis in individuals with biopsy-proven MASH or those at higher risk of clinically significant liver fibrosis identified with noninvasive tests (155–160). Combination therapy with pioglitazone plus GLP-1 RA should also be considered for treatment of hyperglycemia in adults with type 2 diabetes with biopsy-proven MASH or those at higher risk of clinically significant liver fibrosis identified with noninvasive tests, as such therapy is safe and effective and has been shown to reduce hepatic steatosis (161–163). It is important to note that these studies are based on phase 2 clinical trials and that only semaglutide has recently shown benefit in a phase 3 clinical trial with histological outcomes in MASH, including improvements in steatohepatitis and fibrosis (156); this subsequently led to its approval by the FDA for the treatment of MASH with moderate to advanced liver fibrosis, while the other therapies await further phase 3 confirmation of evidence. However, these plans are preferred as they offer potential benefit compared with lack of histological benefit (or clinical trial data) from other glucose-lowering therapies in MASLD. Further details regarding liver health in diabetes can be found in section 4, “Comprehensive Medical Evaluation and Assessment of Comorbidities.”

Basal Insulin

Basal insulin alone is the most convenient initial insulin treatment and can be added to noninsulin glucose-lowering medications. For individuals with type 2 diabetes, starting doses can be estimated based on body weight (0.1–0.2 units/kg/day) and the degree of hyperglycemia, with individualized titration over time as needed to achieve and maintain glycemic goals. The principal action of basal insulin is to restrain hepatic glucose production and limit hyperglycemia overnight and between meals (171,172). Attainment of fasting glucose goals can be achieved with human NPH insulin or a long-acting insulin analog. In clinical trials, long-acting basal analogs (U-100 glargine) have been demonstrated to reduce the risk of level 2 hypoglycemia and nocturnal hypoglycemia compared with NPH insulin (173). Longer-acting basal analogs (U-300 glargine or degludec) convey a lower nocturnal hypoglycemia risk than U-100 glargine (174,175). It is important to understand how to convert individuals from one basal insulin to another, as switching insulins may be required due to the availability of more clinically appropriate insulin alternatives, removal of a product from the market (i.e., insulin detemir), or changes to insurance coverage. Often doses can be converted unit for unit and subsequently adjusted based on glucose monitoring; however, an initial dose reduction of 10–20% can be used for individuals in very tight management or at high risk for hypoglycemia and is typically needed when switching from human NPH insulin or U-300 glargine to another insulin (176). Clinicians should also be aware of the potential for overbasalization with insulin therapy (i.e., use of higher than clinically necessary and appropriate dose of basal insulin, typically masking insufficient).

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mealtime insulin). Clinical signals that should prompt evaluation for overbasalization include high bedtime-to-morning or postprandial-to-preprandial glucose differential (e.g., bedtime-to-morning glucose differential ≥ 50 mg/dL [≥ 2.8 mmol/L]), hypoglycemia (aware or unaware), and high glucose variability. Evidence of over-basalization should prompt reevaluation of the glucose-lowering treatment plan to better address postprandial hyperglycemia (177).

Combination Injectable Therapy and Prandial Insulin

If basal insulin has been titrated to an acceptable fasting blood glucose level and A1C remains above goal, if there is evidence of significant postprandial hyperglycemia, or if signs of overbasalization are present, advancement to combination injectable therapy is necessary (Fig. 9.5). This approach can use a GLP-1 RA or dual GIP and GLP-1 RA added to basal insulin or multiple doses of prandial insulin (131,178). If an individual is not already being treated with a GLP-1 RA or dual GIP and GLP-1 RA, a GLP-1 RA (either as an individual

product or in a fixed-ratio combination with a basal insulin product) or dual GIP and GLP-1 RA should be considered prior to starting prandial insulin to address prandial management and to lower the risks of hypoglycemia and weight gain associated with insulin therapy (131,178).

Further intensification of insulin therapy entails adding doses of prandial insulin to basal insulin. Starting with a single prandial dose with the largest meal of the day is simple and effective, and it can be advanced to a plan with multiple prandial doses if necessary (179). We suggest starting with a prandial insulin dose of 4 units or 10% of the amount of basal insulin at the largest meal or the meal with the greatest postprandial excursion. The prandial insulin plan can then be intensified based on individual needs (Fig. 9.5). Alternatively, for an individual treated with basal insulin in whom additional prandial coverage is desired but administering insulin prior to one or more meals is not feasible, the medication plan can be converted to two doses of a premixed insulin. Each approach has advantages and disadvantages. For example, basal-prandial plans offer greater flexibility for individuals who eat on irregular schedules, have variable meal content, or otherwise benefit from greater individualization and flexibility in insulin administration. On the other hand, two doses of premixed insulin is a simple, convenient means of spreading insulin across the day. Moreover, human insulins, separately, self-mixed, or as premixed NPH/regular (for example, 70/30) formulations, are often less costly alternatives to insulin analogs.

Individuals with type 2 diabetes are generally more insulin resistant than those with type 1 diabetes, require higher daily doses (~1 unit/kg), and have lower rates of hypoglycemia (180). Meta-analyses of trials comparing rapid-acting insulin analogs with human regular insulin in type 2 diabetes have not reported meaningful differences in A1C or hypoglycemia (181). Titration of prandial insulin can be based on home self-monitored blood glucose or CGM. When significant additions to the prandial insulin dose are made, particularly with the evening meal, consideration should be given to decreasing basal insulin to reduce risk of hypoglycemia. When initiating intensification of insulin therapy, metformin, SGLT2 inhibitors, and GLP-1 RAs (or a dual GIP and GLP-1 RA) should be maintained, unless adverse effects (including significant treatment burden) or contraindications are present. Use of sulfonylureas, meglitinides, and DPP-4 inhibitors should be limited or discontinued, as these medications do not have additional beneficial effects on cardiovascular, kidney, weight, or liver outcomes, and sulfonylureas and meglitinides increase risk of hypoglycemia and weight gain. Adjunctive use of pioglitazone may help to improve glycemia and reduce the amount of insulin needed, although potential side effects should be considered and may limit its use.

Alternative Insulin Routes

Insulin is primarily administered via subcutaneous injection or infusion. Administration devices provide some additional variation in the subcutaneous delivery beyond vial and syringe versus insulin pen. Those devices include continuous insulin pumps (programmable or automated basal and bolus settings and fixed basal and bolus settings) and bolus-only insulin patch pump. In addition, prandial or correction insulin doses may be administered using inhaled human insulin. Inhaled insulin is available as monomers of regular human insulin; studies in individuals with type 1 diabetes suggest that inhaled insulin has pharmacokinetics faster than those of RAA (190). Studies comparing inhaled insulin with injectable insulin have demonstrated its faster onset and shorter duration compared with the RAA insulin lispro as well as clinically meaningful A1C reductions.

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ADDITIONAL RECOMMENDATIONS FOR ALL INDIVIDUALS WITH DIABETES

Recommendations

9.24 Include healthy behaviors, diabetes self-management education and support, avoidance of therapeutic inertia, and social determinants of health as essential components of the glucose-lowering management of diabetes. A

9.25 Use of continuous glucose monitoring (CGM) is recommended at diabetes onset and anytime thereafter for adults with diabetes who are on insulin therapy, A on noninsulin therapies that can cause hypoglycemia, B

9.26 Monitor for signs of overbasalization during insulin therapy, such as significant bedtime-to-morning or postprandial-to-preprandial glucose differential, occurrences of hypoglycemia (aware or unaware), and high glycemic variability. When overbasalization is suspected, a thorough reevaluation should occur promptly to further tailor therapy to the individual's needs. E

9.28 Glucagon should be prescribed for all individuals taking insulin or at high risk for hypoglycemia. A Family, caregivers, school personnel, and others providing support to these individuals should know its location and be educated on how to administer it. Glucagon preparations that do not require reconstitution are preferred. B

9.30 In adults with diabetes and cost-related barriers, consider use of lower-cost medications for glycemic management (i.e., metformin, sulfonylureas, thiazolidinediones, and human insulin) within the context of their risks for hypoglycemia, weight gain, cardiovascular and kidney events, and other adverse effects. E

The use of continuous glucose monitoring (CGM) in adults with diabetes improves glycemic outcomes (e.g., A1C, time in range) and reduces episodes of hypoglycemia in adults with diabetes who are on insulin therapy or other noninsulin therapies that can induce hypoglycemia (193–209). In addition, the use of AID systems should be offered to adults with type 1 or type 2 diabetes who are on insulin therapy to improve glycemic management and glycemic outcomes (26,28–32,35–37,210–215).

Costs for noninsulin and insulin diabetes medications have increased dramatically over the past two decades, and an increasing proportion of cost is now passed on to people with diabetes and their families (221). Table 9.3 provides cost information for currently approved noninsulin therapies, while Table 9.4 provides these data for insulin. Of note, prices listed are average wholesale prices (AWP) (222) and National Average Drug Acquisition Costs (NADAC) (223); these estimates allow for a comparison of drug prices but do not represent the actual costs to individuals.

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Class	Compound	Dosage strength/product (if applicable)	Maximum approved daily dose†	Median AWP (min, max)*	Median NADAC (min, max)*

Metformin		Sulfonylureas (1st generation)		Sulfonylureas (2nd generation)		Glimepiride		Glipizide		Glyburide		Thiazolidinedione		Pioglitazone		α-Glucosidase inhibitors		Acarbose		Miglitol		Meglitinides		Nateglinide		Repaglinide		DPP-4 inhibitors		Alogliptin																																	
500 mg (IR)	2,000 mg	\$85 (\$3, \$216)	\$2	500 mg (ER)	2,000 mg	\$89 (\$5, \$6,719)	\$3	850 mg (IR)	2,550 mg	\$108 (\$3, \$189)	\$2	1,000 mg (IR)	2,000 mg	\$87 (\$2, \$144)	\$1	1,000 mg (ER)	2,000 mg	\$1,884 (\$242, \$7,214)	\$26 (\$24, \$28)	500 mg (Sol)	2,000 mg	\$810 (\$810, \$1,478)	\$417	4 mg	8 mg	\$73 (\$71, \$198)	\$2	10 mg (XL/ER)	20 mg	\$48	\$9	6 mg (micronized)	12 mg	\$54 (\$48, \$71)	\$13	5 mg	20 mg	\$88 (\$63, \$432)	\$7	45 mg	45 mg	\$348 (\$7, \$349)	\$4	100 mg	300 mg	\$106 (\$104, \$378)	\$22	100 mg	300 mg	\$294 (\$241, \$346)	\$183	120 mg	360 mg	\$104	\$16	2 mg	16 mg	\$878 (\$799, \$1,728)	\$28	25 mg	25 mg	\$1,000 (\$750, \$1,250)	\$100

| 25 mg | \$234 | \$143 | | • Linagliptin | 5 mg | 5 mg | \$630 | \$504 | | • Saxagliptin | 5 mg | 5 mg | \$524 (\$523, \$524) | \$179 | | • Sitagliptin | 100 mg | 100 mg | \$341 | \$303 | | 25 mg/mL | 100 mg | | \$354 | NA | | SGLT2 inhibitors | • Bexagliflozin | 20 mg | 20 mg | \$47 | NA | | • Canagliflozin | 300 mg | 300 mg | \$718 | \$575 | | • Dapagliflozin | 10 mg | 10 mg | \$664 | \$345 | | • Empagliflozin | 25 mg | 25 mg | \$629 | \$604 | | • Ertugliflozin | 15 mg | 15 mg | \$428 | \$343 | | GLP-1 RAs | • Dulaglutide | 4.5 mg pen | 4.5 mg† | \$1,185 | \$953 | | • Liraglutide | 18 mg/3 mL pen | 1.8 mg | \$929 (\$845, \$929) | \$577 | | • Semaglutide | 2 mg pen | 2 mg† | \$1,197 | \$966 | | 14 mg (tablet) | 14 mg | | \$1,197 | \$965 | | Dual GIP and GLP-1 RA | • Tirzepatide | 15 mg pen | 15 mg† | \$1,296 | \$1,041 | | Bile acid sequestrant | • Colesevelam | 625 mg tabs | 3.75 g | \$692 (\$674, \$712) | \$56 | | 3.75 g suspension | 3.75 g | \$674 (\$673, \$675) | | \$93 | | Dopamine-2 agonist | • Bromocriptine | 0.8 mg | 4.8 mg | \$1,188 | \$957 |

AWP, average wholesale price; DPP-4, dipeptidyl peptidase 4; ER and XL, extended release; GIP, glucose-dependent insulinotropic polypeptide; GLP-1 RA, glucagon-like peptide 1 receptor agonist; IR, immediate release; max, maximum; min, minimum; NA, data not available; NADAC, National Average Drug Acquisition Cost; SGLT2, sodium–glucose cotransporter 2; Sol, solution. AWP and NADAC prices as of 15 July 2025.

*Calculated for 30-day supply (AWP [222] or NADAC [223] unit price × number of doses required to provide maximum approved daily dose × 30 days); median AWP or NADAC listed alone when only one product and/or price. †Used to calculate median AWP and NADAC (min, max); generic prices used, if available commercially. ‡Administered once weekly.

people with diabetes because they do not account for various discounts, rebates, and other price adjustments often involved in prescription sales that affect the actual cost incurred by the individual. Medication costs can be a major source of stress for people with diabetes and contribute to worse medication-taking behavior (224); cost-reducing strategies may improve medication-taking behavior in some cases (225).

Caps on out-of-pocket costs for insulin have been implemented for individuals with Medicare insurance (to \$35 per insulin prescription per month) and individuals with state-regulated commercial insurance plans who live in 26 states and the District of Columbia that implemented such legislation (to either \$35 per insulin prescription per month or \$100 per total monthly insulin payment) (226–228). Additionally, insulin manufacturers have introduced cost reductions and copayment assistance programs; however, these do not cover all insulins, and the copayment assistance programs have variable eligibility requirements and reduce out-of-pocket payments to variable degrees (229) (see section 1, “Improving Care and Promoting Health in Populations”). Individuals with high-deductible health plans and those without insurance coverage can incur very high out-of-pocket expenses for glucose-lowering therapies. Moreover, no such caps exist for diabetes medical equipment (i.e., equipment for glucose monitoring and insulin administration) or for noninsulin medications. It is therefore essential to screen all people with diabetes for financial concerns and cost-related barriers to care and to engage members of the health care team, including

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Table 9.4—Median cost of insulin products in the U.S. calculated as AWP and NADAC per 1,000 units of specified dosage form/product

Insulins	Compounds	Dosage form/product	Median AWP (min, max)*	Median NADAC (min, max)*
		Rapid-acting		
	Aspart	U-100 vial	\$87† \$70†	U-100 cartridge \$107† \$86†
		U-100 prefilled pen	\$112† \$90†	Aspart biosimilars#
		U-100 vial	\$83 NA	U-100 prefilled pen
		U-100 vial	\$347 \$278	U-100 cartridge
		U-100 vial	\$102 \$82	U-100 prefilled pen
	Glulisine	U-100 vial	\$102 \$82	U-100 prefilled pen
		U-100 vial	\$132 \$105	
	Inhaled insulin	Inhalation cartridges	\$1,578 \$1,265	Lispro
		U-100 vial	\$30† \$24†	U-100 cartridge
		U-100 vial	\$123 \$98	U-100 prefilled pen
		U-100 vial	\$127† \$102†	U-200 prefilled pen
		U-100 vial	\$424 \$339	
	Lispro-aabc	U-100 vial	\$330 \$263	U-100 prefilled pen
		U-100 vial	\$424 \$339	U-200 prefilled pen
		U-100 vial	\$118 \$110	U-100 prefilled pen
		U-100 vial	\$151 \$121	
	Short-acting	Human regular	U-100 vial	\$56 (\$54, \$58)‡ \$44 (\$43, \$46)‡
		U-100 vial	\$73	

\$58 | | Intermediate-acting | Human NPH | U-100 vial | \$58 (\$54, \$58)† | \$45 (\$43, \$46)† | | U-100 prefilled pen | \$93 (\$73, \$113) | \$74 (\$58, \$91) | | Concentrated human regular insulin | U-500 human regular insulin | U-500 vial | \$178 | \$143 | | U-500 prefilled pen | | \$230 | \$183 | | Long-acting | Degludec | U-100 vial | \$142† | \$114† | | U-100 prefilled pen | | \$142† | \$114† | | U-200 prefilled pen | | \$142† | \$114† | | Glargine | U-100 vial | \$77 | \$62 | | U-100 prefilled pen | | \$77 | \$62 | | U-300 prefilled pen | | \$152 | \$122† | | Glargine biosimilar/follow-on products | U-100 vial | \$76† | \$61† | | U-100 prefilled pen | \$74 (\$74,† \$261) | \$59 (\$59,† \$209) | | Premixed insulin products | Aspart 70/30 | U-100 vial | \$87†‡ | \$69†‡ | | U-100 prefilled pen | | \$112†‡ | \$90†‡ | | Lispro 50/50 | U-100 vial | \$102 | NA | | U-100 prefilled pen | | \$127 | \$102 | | Lispro 75/25 | U-100 vial | \$102 | \$82 | | U-100 prefilled pen | | \$127† | \$102† | | NPH/regular 70/30 | U-100 vial | \$56 (\$54, \$58) | \$45 (\$43, \$46) | | U-100 prefilled pen | \$93 (\$73, \$113)† | \$74 (\$58, \$90)† | | Premixed insulin/GLP-1 RA products | Degludec/liraglutide | 100/3.6 mg prefilled pen | \$1,073 | \$859 | | Glargine/lixisenatide | 100/33 mg prefilled pen | \$713 | \$571 |

AWP, average wholesale price; GLP-1 RA, glucagon-like peptide 1 receptor agonist; NA, data not available; NADAC, National Average Drug Acquisition Cost. AWP (222) and NADAC (223) prices as of 15 July 2025.

*AWP or NADAC calculated as in Table 9.3. †Unbranded product prices used when available. ‡AWP and NADAC data presented do not include human insulins (approximately \$25/vial or \$43/box of 5 pens) or select analog insulins (approximately \$73/vial or \$86/box of 5 pens) available at Walmart; median listed alone when only one product and/or price. #Pricing for aspart-xjhz not available on 15 July 2025.

SPECIAL CIRCUMSTANCES AND POPULATIONS

Recommendations

1. 9.31a Use of compounded products that are not approved by the U.S. Food and Drug Administration (FDA) is not recommended due to uncertainty about their content and resulting concerns about safety, quality, and effectiveness. 2. 9.31b If a glucose-lowering medication is unavailable (e.g., in shortage), it is.

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recommended to switch to a different FDA-approved medication with similar efficacy, as clinically appropriate. E

9.31c Upon resolution of the unavailability (e.g., shortage), reassess the appropriateness of resuming the original FDA-approved medication. E

9.32a Individuals of childbearing potential with diabetes should be counseled on contraception options A and the impact of some glucose-lowering medications on contraception efficacy. C

9.32b A person-centered shared decision-making approach to preconception planning is essential for all individuals of childbearing potential with diabetes. A Preconception planning should address attainment of glycemic goals, A the time frame for discontinuing noninsulin glucose-lowering medications, E and optimal glycemic management in preparation for pregnancy. A

9.33 Individuals who develop hyperglycemia during treatment with immunotherapy (including anti-PD-1 or anti-PD-L1 therapy, e.g., nivolumab, pembrolizumab, or avelumab) should be assessed for the immediate need for initiation of insulin therapy due to the potential risk of diabetic ketoacidosis while additional testing is completed to determine if the hyperglycemia is related to immunotherapy-associated diabetes. Close monitoring, education, and dose adjustment are needed if insulin is started. C

9.34 Consider metformin as the first-line treatment of hyperglycemia due to mTOR inhibitors. E

9.35a Consider metformin as the first-line treatment of hyperglycemia due to phosphoinositide 3-kinase (PI3K) inhibitors that affect the α isoform (e.g., alpelisib and inavolisib). E

9.35b Use of insulin should be reserved for severe hyperglycemia and hyperglycemic crises due to its potential impact on the efficacy of PI3K inhibitors. E

9.36 Adjust or initiate additional glucose-lowering therapies to maintain individualized glycemic goals based on the specific glucocorticoid treatment plan, with frequent reassessment of glucose levels and glucocorticoid

treatment plans. C

9.37 In adults with posttransplantation diabetes mellitus (PTDM) or preexisting type 2 diabetes, insulin is preferred for the management of hyperglycemia in the postoperative setting. A DPP-4 inhibitor can be considered for mild hyperglycemia. A

9.38a In adults with PTDM or preexisting type 2 diabetes, noninsulin pharmacotherapy can be used for long-term glycemic management, C and medication selection may differ depending on the transplanted organ(s). E

9.38b In adults with PTDM or preexisting type 2 diabetes, a GLP-1 RA can be considered for long-term glycemic management due to additional cardiometabolic benefits (e.g., cardiovascular, kidney, weight, and liver benefits). C

9.38c If long-term individualized glycemic goals cannot be achieved or maintained with noninsulin pharmacotherapy in adults with PTDM or preexisting type 2 diabetes, consider adding insulin. C

9.39 Educate individuals with diabetes who are at risk for developing diabetic ketoacidosis and who are treated with SGLT inhibition on the risks and signs of ketoacidosis and methods of risk mitigation management, provide them with appropriate tools for ketone measurement (i.e., serum β -hydroxybutyrate), and discourage a ketogenic eating pattern. E

Therapeutic Strategies With Medication Unavailability

Health care professionals and people with diabetes struggle when medication supplies are insufficient to meet the demand. Examples of such circumstances include recalls involving a number of metformin products and the marked increase in demand for agents from the GLP-1 RA and dual GIP and GLP-1 RA classes. The latter circumstance led to such a low level of availability that products were determined by the FDA to be in shortage (231). To assist with supply of medications during the time they are in shortage (as signaled by their inclusion on the FDA Drug Shortages Database), compounding pharmacies and outsourcing compounding facilities are allowed to make copies, or products that are essentially duplicates of the marketed FDA-approved product (232). A significant number of concerning reports regarding safety and efficacy of compounded incretin products have emerged, including using active ingredients that are not proven safe or effective for use in humans, incorporation of additional ingredients not clinically tested when mixed with incretin products (e.g., vitamin B12 and vitamin B6), products provided in nonstandard concentrations and doses and/or multidose vials and prefilled syringes not accompanied by education or labeling to mitigate administration errors, and the emergence of counterfeit products that pose significant risk to individuals taking these products (233–236). Due to safety, quality, and effectiveness concerns, use of non-FDA-approved compounded products is not recommended (237). Instead, consider switching to a different FDA-approved medication as clinically appropriate (238). Once the desired FDA-approved product becomes available, individuals should be reassessed to determine the appropriateness of resuming the product based on their current care needs, preferences, and priorities.

Care Considerations for Individuals of Childbearing Potential

The impact of glycemia during pregnancy is well understood; however, evidence for the safe use of noninsulin glucose-lowering medications is limited (see section 15, “Management of Diabetes in Pregnancy”). Studies on the efficacy and safety of glucose-lowering medications exclude individuals who are pregnant and require individuals of childbearing potential to use one or two forms of contraception. It is recommended that individuals of childbearing potential use a form of contraception when also taking glucose-lowering medications with unknown risks, limited evidence on safety, or known risks during pregnancy, regardless of the individual’s intention to become pregnant, as many pregnancies are unplanned. The options for contraception should be discussed with all individuals of childbearing potential with diabetes and should include information regarding the potential impact of glucose-lowering medications on the effectiveness of contraception. Medications that affect gastrointestinal emptying time (e.g., GLP-1 RAs or dual GIP and GLP-1 RA) may affect the absorption of orally administered medications, including oral contraception. The impact on gastric emptying with GLP-1 RAs and the dual GIP and GLP-1 RA is highest at initiation and with dosage increases and then diminishes with continued administration (239). Tirzepatide, the

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Dual GIP and GLP-1 RA, was shown to affect the levels of oral contraception during the time of its highest impact on gastric emptying; the GLP-1 RAs may affect the levels of oral contraception as well but to a lesser extent than tirzepatide (240,241). Thus, individuals starting or increasing doses of tirzepatide who also take oral contraception should use a second form of contraception until the maintenance dose of tirzepatide is achieved and used for at least 4 weeks (242).

Preconception counseling should be part of the routine care of individuals with diabetes who have childbearing potential. Counseling should include the known benefits and risks of glucose-lowering medications as well as other medications (e.g., lipid-lowering and anti-hypertensive therapies) during pregnancy and recommendations for when changes in medications should occur prior to pregnancy. Individuals planning pregnancy should be counseled that a period of several months is usually needed and adjustment of therapy approved for use in pregnancy to achieve preconception glycemic goals prior to pregnancy (see section 15, "Management of Diabetes in Pregnancy," for more information on preconception counseling and glucose-lowering treatment during pregnancy).

Therapeutic Strategies for Individuals Receiving Cancer Treatment

Hyperglycemia due to chemotherapy may either be transient (improving upon treatment cessation) or represent permanent diabetes. Immune checkpoint inhibitors (ICIs) impair regulatory components of the immune system, allowing for immunogenic response against cancer cells, which can result in autoimmune toxicities, including an autoimmune form of diabetes that results in β -cell destruction (incidence approximately $\leq 1\%$) (243–246). This form of diabetes is most common after exposure to ICIs that target programmed cell death protein 1 (PD-1) (i.e., nivolumab and pembrolizumab) and those that target programmed cell death protein ligand 1 (PD-L1) (i.e., durvalumab and avelumab). ICIs that target cytotoxic T-lymphocyte–associated protein 4 (CTLA-4) (i.e., ipilimumab) have also been implicated in this process, but much less commonly (247).

Hyperglycemia as a result of ICIs can occur at any time after the initiation of therapy—as quickly as 1 week after the first dose to up to 12 months after. Insulin therapy is the cornerstone of management, as individuals typically present with rapid-onset, severe hyperglycemia or DKA (248). Early initiation of therapy can prevent these drastic presentations, and the initiation of basal insulin should be considered in individuals with blood glucose >250 mg/dL while further evaluation takes place. Prandial insulin is often required as well, if insulinopenia is confirmed. ICI treatment should not be discontinued in the event of severe hyperglycemia, as the β -cell destruction associated with this process is irreversible. Lifelong insulin therapy is generally required (249).

Metformin is the first-line treatment of hyperglycemia secondary to mTOR inhibitor treatment because of its efficacy and safety profile. Due to its ability to reduce insulin resistance, pioglitazone may be considered as a second-line treatment, depending on the risks of its adverse effects to the individual. There is no direct evidence regarding the efficacy of GLP-1 RAs or SGLT2 inhibitors for mTOR inhibitor–induced hyperglycemia; however, there is also no evidence that they impair the efficacy of the mTOR inhibitor. Thus, evaluation of their use as second- or third-line treatments for this circumstance should be made based on their overall benefits and risks. Insulin is typically reserved for cases of refractory hyperglycemia after noninsulin treatments are used, in the presence of intolerance or contraindications to noninsulin treatments, or for severe hyperglycemia (260–262).

Glucocorticoids (including prednisone and dexamethasone), which are often used as part of acute, adjunctive, and chronic treatment of cancer and inflammatory conditions (e.g., rheumatoid arthritis and inflammatory bowel disease) as well as posttransplantation, cause hyperglycemia primarily by increased insulin resistance and hepatic glucose production. Other contributing effects include increased appetite, decreased production/secretion of insulin, and enhanced effects of counterregulatory hormones (such as epinephrine) (263–266). The timing and extent of hyperglycemia vary based on the dose, duration, route of administration (i.e., intravenous, oral, or intraarticular) and the specific glucocorticoid used (265–271). For example, with

morning administration of prednisone (with the first meal of the day), glucose starts rising after the first meal, with peak effects in the afternoon and evening, and declines to baseline by the next morning; in contrast, with a single dose of dexamethasone, glucose elevations may last more than 24 h with some decline overnight. The extent of the elevation is dependent on the dose, so as the glucocorticoid is tapered, the extent of hyperglycemia will.

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Monitoring glucose levels solely in the morning may miss the extent of the hyperglycemia experienced due to glucocorticoid use and prevent appropriate management (263,265–267,272). Thus, glucose-lowering medication adjustments or additions should match the timing and extent of hyperglycemia and allow for rapid adjustment as the dose of the glucocorticoid dose changes to minimize the likelihood of hyperglycemia and hypoglycemia. Insulin is the most frequently used glucose-lowering medication to manage hyperglycemia secondary to glucocorticoid use. The selection of insulin type and dose depends on the dose and duration of the glucocorticoid (265,266,269,272–275). Additions or dose adjustments of sulfonylureas, including meglitinides, have also been used for those with type 2 diabetes or no previous diagnosis of diabetes. Due to lack of direct evidence and the time needed to achieve the glucose-lowering effect, additions or dose adjustments of other glucose-lowering medications are reserved for when stable doses of glucocorticoids are chronically administered and are not for acute management (265,266,269,272, 275,276).

Pancreatic Diabetes and Cystic Fibrosis–Related Diabetes

Individuals with pancreatic diabetes may require early insulin initiation to achieve and maintain glycemic goals. In individuals with a history of pancreatitis, use of incretin medications (i.e., GLP-1 RAs, a dual GIP and GLP-1 RA, and DPP-4 inhibitors) should be avoided (see section 2, “Diagnosis and Classification of Diabetes”). Individuals with cystic fibrosis–related diabetes should be treated with insulin therapy; insulin pump therapy, including AID systems, should be considered when appropriate (277).

Posttransplantation Diabetes Mellitus

The diagnosis of posttransplantation diabetes mellitus (PTDM) relies on the same glycemic characteristics as other forms of diabetes. However, due to the unique effects of immunosuppressant drugs, an oral glucose tolerance test (OGTT) is the preferred and most accurate method for diagnosis (compared with A1C or fasting plasma glucose) (278). In 2024, an update from the 3rd International PTDM Consensus Meeting recommended screening for diabetes risk factors with a pretransplant OGTT while the individual is on the waitlist, followed by early OGTT at 3 months posttransplantation to diagnose PTDM and a late OGTT at 1 year and onward as appropriate (279).

Metformin can be used but with caution; it should not be initiated if eGFR is ≤ 45 mL/min/1.73 m², and it should be discontinued with eGFR ≤ 30 mL/min/1.73 m². Limitations in metformin use include the risk of lactic acidosis with fluctuating kidney function, either with graft dysfunction or rejection in kidney transplantation or acute kidney injury in other solid organ transplantation. Metformin use may be associated with lower risks of cardiac allograft vasculopathy after heart transplantation (288) and all-cause, malignancy-related, and infection-related mortality after kidney transplantation (289).

DPP-4 inhibitors have been shown to be safe and effective posttransplantation in RCTs, even in the immediate posttransplant period for mild hyperglycemia or impaired glucose tolerance, and have the potential to decrease progression to PTDM (290). GLP-1 RA therapy may be preferred for many individuals, as shown by increasing evidence from large retrospective studies on the benefits of GLP-1 RAs on cardiovascular, kidney, weight loss, and glucose lowering outcomes. Studies have not found any concern for negative interaction with immunosuppressants, which would necessitate changes in dosing (291–303). However, caution should be used when gastrointestinal side effects occur, particularly if individuals have this additive effect to the side effects of the immunosuppressants. Additionally, in lung transplant recipients, gastroparesis and gastroesophageal reflux are of particular concern, as these conditions may induce allograft lung damage and are frequent side effects of GLP-1 RAs. A gastric emptying study may be useful in identifying ideal candidates before initiation.

SGLT2 inhibitors may be similarly preferred for individuals with ASCVD, HF, and CKD and appear to be safe and effective in PTDM (304–309). However, there is increased risk of genitourinary tract infection, which is of particular concern in immunosuppressed individuals. Of note, kidney transplant recipients have an innate anatomical increased risk of urinary tract infections (UTIs), particularly immediately posttransplant when the ureteral stent is still in place. Prior history of UTIs and individual risk of UTIs should be considered when using this class after kidney transplant.

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It is essential that diabetes management posttransplantation is implemented in the setting of an interprofessional team. The transplant physicians and the endocrinologists should work collaboratively to monitor for medication toxicity and deleterious side effects that may affect allograft function and to optimize cardiovascular and metabolic outcomes.

Maturity-Onset Diabetes of the Young

Individuals with maturity-onset diabetes of the young due to HNF1A and HNF4A mutations can be treated with low-dose sulfonylurea therapy but may ultimately require insulin therapy (310) (see section 2, “Diagnosis and Classification of Diabetes”) (Table 2.7). For those with HNF1A mutations, addition of a DPP-4 inhibitor to the sulfonylurea may help improve glycemic variability and attainment of glycemic goals (311). Individuals with neonatal diabetes due to KCNJ22 and ABCC8 mutations can be treated with high-dose sulfonylureas, while those with INS, GATA6, EIF2AK3, and FOXP3 mutations require insulin therapy (310).

SGLT Inhibition and Risk of Ketosis

Individuals with type 1 diabetes (83,312) and insulin-deficient type 2 diabetes are at increased risk for DKA with SGLT inhibitor therapy (SGLT2 inhibitors or the SGLT1/2 inhibitor). SGLT inhibitor–associated DKA occurs in approximately 4% of people with type 1 diabetes; the risk can be 5–17 times higher than that in people with type 1 diabetes not treated with SGLT inhibitors (313). It is important to note that SGLT2 inhibitors are not approved for use in people with type 1 diabetes. In contrast, DKA is uncommon in people with type 2 diabetes treated with SGLT inhibitors, with an estimated incidence of 0.6–4.9 events per 1,000 person-years (314). Risk factors for DKA in individuals with either type 1 or type 2 diabetes treated with SGLT inhibitors include very-low-carbohydrate eating patterns, prolonged fasting, dehydration, excessive alcohol intake, and other common precipitating factors (83,312). Up to a third of people treated with SGLT2 inhibitors who developed DKA present with glucose levels ≥ 200 mg/dL (≥ 11.1 mmol/L) (315), and in one study 71% presented with glucose levels ≤ 250 mg/dL (≤ 13.9 mmol/L) (316); therefore, it is important to educate at-risk individuals about the signs and symptoms of DKA and DKA mitigation.

References

1. Diabetes Control and Complications Trial (DCCT)/Epidemiology of Diabetes Interventions and Complications (EDIC) Study Research Group. Mortality in type 1 diabetes in the DCCT/EDIC versus the general population. *Diabetes Care* 2016;39:1378–1383
2. Lachin JM, Bebu I, Nathan DM; DCCT/EDIC Research Group. The beneficial effects of earlier versus later implementation of intensive therapy in type 1 diabetes. *Diabetes Care* 2021;44:2225–2230
3. Lachin JM, Nathan DM; DCCT/EDIC Research Group. Understanding metabolic memory: the prolonged influence of glycemia during the Diabetes Control and Complications Trial (DCCT) on future risks of complications during the study of the Epidemiology of Diabetes Interventions and Complications (EDIC). *Diabetes Care* 2021;44:2216–2224
4. Holt RIG, DeVries JH, Hess-Fischl A, et al. The management of type 1 diabetes in adults. a consensus report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). *Diabetes Care* 2021;44:2589–2625
5. Tricco AC, Ashoor HM, Antony J, et al. Safety, effectiveness, and cost effectiveness of long acting versus intermediate acting insulin for patients with type 1 diabetes: systematic review and network meta-analysis. *BMJ* 2014;349:g5459
6. Bartley PC, Bogoev M, Larsen J, Philotheou A. Long-term efficacy and safety of insulin detemir compared to Neutral Protamine Hagedorn insulin in patients with type 1 diabetes using a treat-to-target basal-bolus regimen with

insulin aspart at meals: a 2-year, randomized, controlled trial. *Diabet Med* 2008;25:442–449 7. Little S, Shaw J, Home P. Hypoglycemia rates with basal insulin analogs. *Diabetes Technol Ther* 2011;13(Suppl. 1):S53–S64 8. Aronson R, Biester T, Leohr J, et al. Ultra rapid lispro showed greater reduction in postprandial glucose versus Humalog in children, adolescents and adults with type 1 diabetes mellitus. *Diabetes Obes Metab* 2023;25:1964–1972 9. Heise T, Pieber TR, Danne T, Erichsen L, Haahr H. A pooled analysis of clinical pharmacology trials investigating the pharmacokinetic and pharmacodynamic characteristics of fast-acting insulin aspart in adults with type 1 diabetes. *Clin Pharmacokinet* 2017;56:551–559 10. Bode BW, McGill JB, Lorber DL, Gross JL, Chang PC, Bregman DB; Affinity 1 Study Group. Inhaled Technosphere insulin compared with injected prandial insulin in type 1 diabetes: a randomized 24-week trial. *Diabetes Care* 2015;38:2266–2273 11. Russell-Jones D, Bode BW, De Block C, et al. Fast-acting insulin aspart improves glycemic control in basal-bolus treatment for type 1 diabetes: results of a 26-week multicenter, active-controlled, trial. *N Engl J Med* 2019;381:1707–1717

diabetesjournals.org/care # Pharmacologic Approaches to Glycemic Treatment

S209

MiniMed advanced hybrid closed-loop delivery: results from a randomized crossover trial comparing automated insulin delivery with predictive low glucose suspend in people with type 1 diabetes. *Diabetes Care* 2021;44:969–975

40. Mitsui Y, Kuroda A, Ishizu M, et al. Basal insulin requirement in patients with type 1 diabetes depends on the age and body mass index. *J Diabetes Investig* 2022;13:292–298

41. Castellano E, Attanasio R, Giagulli VA, et al.; Associazione Medici Endocrinologi (AME). The basal to total insulin ratio in outpatients with diabetes on basal-bolus regimen. *J Diabetes Metab Disord* 2018;17:393–399

42. Matejko B, Kukułka A, Kiec-Wilk B, Stańczak A, Klupa T, Malecki MT. Basal insulin dose in adults with type 1 diabetes mellitus on insulin pumps in real-life clinical practice: a single-center experience. *Adv Med* 2018;2018:1473160

43. King AB. Mean basal insulin dose is 0.2 U/kg/d at near normal glycaemia for type 1 or 2 diabetes on continuous subcutaneous insulin infusion or once-nightly basal insulin. *Diabetes Obesity Metabolism* 2021;23:866–869

44. Peters AL, Laffel L. The American Diabetes Association/JDRF Type 1 Diabetes Sourcebook. American Diabetes Association, 2013

45. Chiang JL, Kirkman MS, Laffel LMB, Peters AL, Type 1 Diabetes Sourcebook Authors. Type 1 diabetes through the life span: a position statement of the American Diabetes Association. *Diabetes Care* 2014;37:2034–2054

46. Samann A, Muhlhauser I, Bender R, Hunger-Dathe W, Kloos C, Muller UA. Flexible intensive insulin therapy in adults with type 1 diabetes and high risk for severe hypoglycemia and diabetic ketoacidosis. *Diabetes Care* 2006;29:2196–2199

47. Builes-Montano CE, Ortiz-Cano NA, Ramirez-Rincón A, Rojas-Henao NA. Efficacy and safety of carbohydrate counting versus other forms of dietary advice in patients with type 1 diabetes mellitus: a systematic review and meta-analysis of randomised clinical trials. *J Hum Nutr Diet* 2022;35:1030–1042

48. Al Balwi R, Al Madani W, Al Ghamdi A. Efficacy of insulin dosing algorithms for high-fat high-protein mixed meals to control postprandial glycemic excursions in people living with type 1 diabetes: a systematic review and meta-analysis. *Pediatr Diabetes* 2022;23:1635–1646

49. DAFNE Study Group. Training in flexible, intensive insulin management to enable dietary freedom in people with type 1 diabetes: Dose Adjustment For Normal Eating (DAFNE) randomised controlled trial. *BMJ* 2002;325:746

50. Hopkins D, Lawrence I, Mansell P, et al. Improved biomedical and psychological outcomes 1 year after structured education in flexible insulin therapy for people with type 1 diabetes: the U.K. DAFNE experience. *Diabetes Care* 2012;35:1638–1642

51. Speight J, Amiel SA, Bradley C, et al. Long-term biomedical and psychosocial outcomes following DAFNE (Dose Adjustment For Normal Eating) structured education to promote intensive insulin therapy in adults with sub-optimally controlled type 1 diabetes. *Diabetes Res Clin Pract* 2010;89:22–29
52. Bruttomesso D, Boscari F, Lepore G, et al. A “slide rule” to adjust insulin dose using trend arrows in adults with type 1 diabetes: test in silico and in real life. *Diabetes Ther* 2021;12:1313–1324
53. Aleppo G, Laffel LM, Ahmann AJ, et al. A practical approach to using trend arrows on the Dexcom G5 CGM system for the management of adults with diabetes. *J Endocr Soc* 2017;1:1445–1460
54. Buckingham B, Xing D, Weinzimer S, et al.; Diabetes Research In Children Network (DirecNet) Study Group. Use of the DirecNet Applied Treatment Algorithm in children with type 1 diabetes. *Diabetes Care* 2017;40:1–8
55. Parise M, Di Molfetta S, Graziano RT, et al. A head-to-head comparison of two algorithms for adjusting mealtime insulin doses based on CGM trend arrows in adult patients with type 1 diabetes: results from an exploratory study. *Int J Environ Res Public Health* 2023;20:3945
56. Petrovski G, Campbell J, Pasha M, et al. Simplified meal announcement versus precise carbohydrate counting in adolescents with type 1 diabetes using the MiniMed 780G advanced hybrid closed loop system: a randomized controlled trial comparing glucose control. *Diabetes Care* 2023;46:544–550
57. Turrin KB, Trujillo JM. Effects of diabetes numeracy on glycemic control and diabetes self-management behaviors in patients on insulin pump therapy. *Diabetes Ther* 2019;10:1337–1346
58. White RO, Wolff K, Cavanaugh KL, Rothman R. Addressing health literacy and numeracy to improve diabetes education and care. *Diabetes Spectr* 2010;23:238–243
59. Sacks DB, Arnold M, Bakris GL, et al. Guidelines and recommendations for laboratory analysis in the diagnosis and management of diabetes mellitus. *Diabetes Care* 2023;46:e151–e199
60. Klonoff DC, Berard L, Franco DR, et al. Advance insulin injection technique and education with FITTER forward expert recommendations. *Mayo Clin Proc* 2025;100:682–699
61. Bergenstal RM, Strock ES, Peremislov D, Gibney MA, Parvu V, Hirsch LJ. Safety and efficacy of insulin therapy delivered via a 4mm pen needle in obese patients with diabetes. *Mayo Clin Proc* 2015;90:329–338
62. Qiao Y-C, Ling W, Pan Y-H, et al. Efficacy and safety of pramlintide injection adjunct to insulin therapy in patients with type 1 diabetes mellitus: a systematic review and meta-analysis. *Oncotarget* 2017;8:66504–66515
63. Meng H, Zhang A, Liang Y, Hao J, Zhang X, Lu J. Effect of metformin on glycaemic control in patients with type 1 diabetes: a meta-analysis of randomized controlled trials. *Diabetes Metab Res Rev* 2018;34:e2983
64. Petrie JR, Chaturvedi N, Ford I, et al.; REMOVAL Study Group. Cardiovascular and metabolic effects of metformin in patients with type 1 diabetes (REMOVAL): a double-blind, randomised, placebo-controlled trial. *Lancet Diabetes Endocrinol* 2017;5:597–609
65. Mathieu C, Zinman B, Hemmingsson JU, et al.; ADJUNCT ONE Investigators. Efficacy and safety of liraglutide added to insulin treatment in type 1 diabetes: the ADJUNCT ONE treat-to-target randomized trial. *Diabetes Care* 2016;39:1702–1710
66. Åhrén B, Hirsch IB, Pieber TR, et al.; ADJUNCT TWO Investigators. Efficacy and safety of liraglutide added to capped insulin treatment in subjects with type 1 diabetes: the ADJUNCT TWO randomized trial. *Diabetes Care* 2016;39:1693–1701
67. von Herrath M, Bain SC, Bode B, et al.; Anti-IL-21–liraglutide Study Group investigators and contributors. Anti-interleukin-21 antibody and liraglutide for the preservation of β -cell function in adults with recent-onset type 1 diabetes: a randomised, double-blind, placebo-controlled, phase 2 trial. *Lancet Diabetes Endocrinol* 2021;9:212–224

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http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S183/848878/dc26s009.pdf

weight and obese patients with type 1 diabetes. with diabetes and recent worsening heart failure.

| 97. | Maruthur NM, Tseng E, Hutfless S, et al. | Diabetes Technol Ther 2024;26:184–189 ||| --- | -----

-----	-----
----- || N Engl J Med
2021;384:117–128 | Diabetes medications as monotherapy or metformin-based combination therapy for type 2
diabetes: a systematic review and meta-analysis. Ann Intern Med 2016;164:740–751 ||| 69. | Garg SK, Akturk
HK, Kaur G, Beatson C, Snell-Bergeon J. Efficacy and safety of tirzepatide in overweight and obese adult
patients with type 1 diabetes. Diabetes Technol Ther 2024;26:367–374 ||| 70. | Pasqua M-R, Tsoukas MA,
Kobayati A, Aboznadah W, Jafar A, Haidar A. Subcutaneous weekly semaglutide with automated insulin
delivery in type 1 diabetes: a double-blind, randomized, crossover trial. Nat Med 2025;31:1239–1245 ||| 71. |
Akturk HK, Dong F, Snell-Bergeon JK, Karakus KE, Shah VN. Efficacy and safety of tirzepatide in adults with
type 1 diabetes: a proof of concept observational study. J Diabetes Sci Technol 2025;19:292–296 ||| 72. |
Shah VN, Akturk HK, Kruger D, et al. Semaglutide in adults with type 1 diabetes and obesity. NEJM Evid
2025;4:EVIDoa2500173 ||| 73. | National Library of Medicine. National Center for Biotechnology Information.
ClinicalTrials.gov. Type 1 Diabetes Impacts of Semaglutide on Cardiovascular Outcomes (T1-DISCO)
(NCT05819138). Accessed 23 August 2025. Available from
<https://clinicaltrials.gov/study/NCT05819138?tab=results> ||| 74. | National Library of Medicine. National
Center for Biotechnology Information. ClinicalTrials.gov. Trial of Semaglutide for Diabetic Kidney Disease in
Type 1 Diabetes (RT1D) (NCT05822609). Accessed 20 August 2025. Available from
<https://www.clinicaltrials.gov/study/NCT05822609> ||| 75. | National Library of Medicine. National Center for
Biotechnology Information. ClinicalTrials.gov. Alleviating Carbohydrate Counting for Patients with Type-1
Diabetes Using a Closed Loop System with Weekly Subcutaneous Semaglutide (SEMA SMA) (NCT06387199).
Accessed 20 August 2025. Available from <https://clinicaltrials.gov/study/NCT06387199> ||| 76. | National
Library of Medicine. National Center for Biotechnology Information. ClinicalTrials.gov. Obesity Complicating
Type 1 Diabetes: GLP-1 Analogue Anti-obesity Treatment (NCT06411210). Accessed 20 August 2025.
Available from <https://clinicaltrials.gov/study/NCT06411210> ||| 77. | Rao L, Ren C, Luo S, Huang C, Li X.
Sodium-glucose cotransporter 2 inhibitors as an add-on therapy to insulin for type 1 diabetes mellitus:
meta-analysis of randomized controlled trials. Acta Diabetol 2021;58:869–880 ||| 78. | Li M, Liu Z, Yang X, et
al. The effect of sodium-glucose cotransporter 2 inhibitors as an adjunct to insulin in patients with type 1
diabetes assessed by continuous glucose monitoring: a systematic review and meta-analysis. J Diabetes
Complications 2023;37:108632 ||| 79. | Chen M-B, Xu R-J, Zheng Q-H, Zheng X-W, Wang H. Efficacy and
safety of sotagliflozin adjuvant therapy for type 1 diabetes mellitus: a systematic review and meta-analysis.
Medicine (Baltimore) 2020;99:e20875 ||| 80. | U.S. Food and Drug Administration. FDA Introductory
Remarks: January 17, 2019: Endocrinologic and Metabolic Drugs Advisory Committee Meeting. 2019.
Accessed 25 August 2025. Available from
<https://wayback.archive-it.org/7993/20190207212714/https://www.fda.gov/> ||| 81. | Bhatt DL, Szarek M,
Steg PG, et al.; SOLOIST-WHF Trial Investigators. Sotagliflozin in patients with diabetes and recent worsening
heart failure. N Engl J Med 2021;384:117–128 ||| 82. | Dan T, Garg S, Peters AL, et al. International
consensus on risk management of diabetic ketoacidosis in patients with type 1 diabetes treated with
sodium-glucose cotransporter (SGLT) inhibitors. Diabetes Care 2019;42:1147–1154 ||| 83. | Herold KC,
Bundy BN, Long SA, et al.; Type 1 Diabetes TrialNet Study Group. An anti-CD3 antibody, teplizumab, in
relatives at risk for type 1 diabetes. N Engl J Med 2019;381:603–613 ||| 84. | Lachin JM, McGee P, Palmer
JP; DCCT/EDIC Research Group. Impact of C-peptide preservation on metabolic and clinical outcomes in the
Diabetes Control and Complications Trial. Diabetes 2014;63:739–748 ||| 85. | Kosheleva L, Koshelev D,
Lagunas-Rangel FA, Levit S, Rabinovitch A, Schioth HB. Disease-modifying pharmacological treatments of
type 1 diabetes: molecular mechanisms, target checkpoints, and possible combinatorial treatments. Pharmacol
Rev 2025;77:100044 ||| 86. | Dean PG, Kukla A, Stegall MD, Kudva YC. Pancreas transplantation. BMJ
2017;357:j1321 ||| 87. | U.S. Food & Drug Administration. Lantidra. Accessed 25 August 2025. Available from

<https://www.fda.gov/vaccines-blood-biologics/lantidra> ||| 88. | Geta ET, Terefa DR, Hailu WB, et al. Effectiveness of shared decision-making for glycaemic control among type 2 diabetes mellitus adult patients: a systematic review and meta-analysis. *PLoS One* 2024;19:e0306296 ||| 89. | Davies MJ, Aroda VR, Collins BS, et al. Management of hyperglycemia in type 2 diabetes, 2022. A consensus report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). *Diabetes Care* 2022;45:2753–2786 ||| 90. | Lingvay I, Sumithran P, Cohen RV, Le Roux CW. Obesity management as a primary treatment goal for type 2 diabetes: time to reframe the conversation. *Lancet* 2022;399:394–405 ||| 91. | Wing RR, Lang W, Wadden TA, et al.; Look AHEAD Research Group. Benefits of modest weight loss in improving cardiovascular risk factors in overweight and obese individuals with type 2 diabetes. *Diabetes Care* 2011;34:1481–1486 ||| 92. | Adler AI, Coleman RL, Leal J, Whiteley WN, Clarke P, Holman RR. Post-trial monitoring of a randomised controlled trial of intensive glycaemic control in type 2 diabetes extended from 10 years to 24 years (UKPDS 91). *Lancet* 2024;404:145–155 ||| 93. | Kunutsor SK, Balasubramanian VG, Zaccardi F, et al. Glycaemic control and macrovascular and microvascular outcomes: a systematic review and meta-analysis of trials investigating intensive glucose-lowering strategies in people with type 2 diabetes. *Diabetes Obes Metab* 2024;26:2069–2081 ||| 94. | Holman RR, Paul SK, Bethel MA, Matthews DR, Neil HAW. 10-year follow-up of intensive glucose control in type 2 diabetes. *N Engl J Med* 2008;359:1577–1589 ||| 95. | Bahardoust M, Mousavi S, Yariali M, et al. Effect of metformin (vs. placebo or sulfonylurea) on all-cause and cardiovascular mortality and incident cardiovascular events in patients with diabetes: an umbrella review of systematic reviews with meta-analysis. *J Diabetes Metab Disord* 2024;23:27–38 ||| 96. | Babu A, Mehta A, Guerrero P, et al. Safe and simple emergency department discharge therapy for patients with type 2 diabetes mellitus and severe hyperglycemia. *Endocr Pract* 2009;15:696–704 ||| 97. | Buse JB, Peters A, Russell-Jones D, et al. Is insulin the most effective injectable antihyperglycaemic therapy? *Diabetes Obes Metab* 2015;17:145–151 ||| 98. | D'Alessio D, Haring H-U, Charbonnel B, et al.; EAGLE Investigators. Comparison of insulin glargine and liraglutide added to oral agents in patients with poorly controlled type 2 diabetes. *Diabetes Obes Metab* 2015;17:170–178 |

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S211

1. Cahn A, Cefalu WT. Clinical considerations for use of initial combination therapy in type 2 diabetes. *Diabetes Care* 2016;39(Suppl. 2):S137–S145 2. Abdul-Ghani MA, Puckett C, Triplitt C, et al. Initial combination therapy with metformin, pioglitazone and exenatide is more effective than sequential add-on therapy in subjects with new-onset diabetes. Results from the Efficacy and Durability of Initial Combination Therapy for Type 2 Diabetes (EDICT): a randomized trial. *Diabetes Obes Metab* 2015;17:268–275 3. Cai X, Gao X, Yang W, Han X, Ji L. Efficacy and safety of initial combination therapy in treatment-naïve type 2 diabetes patients: a systematic review and meta-analysis. *Diabetes Ther* 2018;9:1995–2014 4. Aroda VR, González-Galvez G, Grøn R, et al. Durability of insulin degludec plus liraglutide versus insulin glargine U100 as initial injectable therapy in type 2 diabetes (DUAL VIII): a multicentre, open-label, phase 3b, randomised controlled trial. *Lancet Diabetes Endocrinol* 2019;7:596–605 5. Khunti K, Davies MJ. Clinical inertia-time to reappraise the terminology? *Prim Care Diabetes* 2017;11:105–106 6. Bennett WL, Maruthur NM, Singh S, et al. Comparative effectiveness and safety of medications for type 2 diabetes: an update including new drugs and 2-drug combinations. *Ann Intern Med* 2011;154:602–613 7. Maloney A, Rosenstock J, Fonseca V. A model-based meta-analysis of 24 antihyperglycemic drugs for type 2 diabetes: comparison of treatment effects at therapeutic doses. *Clin Pharmacol Ther* 2019;105:1213–1223 8. Lajthia E, Bucheit JD, Nadpara PA, et al. Combination therapy with once-weekly glucagon like peptide-1 receptor agonists and dipeptidyl peptidase-4 inhibitors in type 2 diabetes: a case series. *Pharm Pract (Granada)* 2019;17:1588 9. Violante R, Oliveira JHA, Yoon K-H, et al. A randomized non-inferiority study comparing the addition of exenatide twice daily to sitagliptin or switching from sitagliptin to exenatide twice daily in patients with type 2 diabetes experiencing inadequate glycaemic control on metformin and sitagliptin. *Diabet Med* 2012;29:e417–e424 10. Nauck MA, Kahle M, Baranov O, Deacon CF, Holst JJ. Addition of a dipeptidyl peptidase-4 inhibitor, sitagliptin, to ongoing therapy with the glucagon-like peptide-1 receptor agonist liraglutide: a randomized controlled trial in patients with type 2 diabetes. *Diabetes Obes Metab* 2017;19:200–207 11. Pratley R, Amod A, Hoff ST, et al.; PIONEER 4 investigators. Oral semaglutide versus subcutaneous liraglutide and placebo in type 2 diabetes (PIONEER 4): a randomised, double-blind, phase 3a

trial. *Lancet* 2019;394:39–50 12. Del Prato S, Kahn SE, Pavo I, et al.; SURPASS-4 Investigators. Tirzepatide versus insulin glargine in type 2 diabetes and increased cardiovascular risk (SURPASS-4): a randomised, open-label, parallel-group, multicentre, phase 3 trial. *Lancet* 2021;398:1811–1824 13. Singh S, Wright EE, Kwan AYM, et al. Glucagon-like peptide-1 receptor agonists compared with basal insulins for the treatment of type 2 diabetes mellitus: a systematic review and meta-analysis. *Diabetes Obes Metab* 2017;19:228–238 14. Levin PA, Nguyen H, Wittbrodt ET, Kim SC. Glucagon-like peptide-1 receptor agonists: a systematic review of comparative effectiveness research. *Diabetes Metab Syndr Obes* 2017;10:123–139

2026 January 29 on guest by

http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S183/848878/dc26s009.pdf from Downloaded

Pharmacologic Approaches to Glycemic Treatment

Diabetes Care Volume 49, Supplement 1, January 2026

type 2 diabetes: the GRADE randomized clinical trial.

| 168. Ahmann AJ, Capehorn M, Charpentier G, et al. | *JAMA Intern Med* 2023;183:705–714 ||

----- | -----
----- | | 155. Belfort R, Harrison SA, Brown K, et al. | A placebo-controlled trial of pioglitazone in subjects with nonalcoholic steatohepatitis. *N Engl J Med* 2006;355:2297–2307 | | 156. Sanyal AJ, Newsome PN, Kliers I, et al.; ESSENCE Study Group. | Phase 3 trial of semaglutide in metabolic dysfunction-associated steatohepatitis. *N Engl J Med* 2025;392:2089–2099 | | 157. Newsome PN, Buchholtz K, Cusi K, et al.; NN9931-4296 Investigators. | A placebo-controlled trial of subcutaneous semaglutide in nonalcoholic steatohepatitis. *N Engl J Med* 2021;384:1113–1124 | | 158. Loomba R, Hartman ML, Lawitz EJ, et al.; SYNERGY-NASH Investigators. | Tirzepatide for metabolic dysfunction-associated steatohepatitis with liver fibrosis. *N Engl J Med* 2024;391:299–310 | | 159. Gastaldelli A, Cusi K, Fernández Landó L, Bray R, Brouwers B, Rodríguez A. | Effect of tirzepatide versus insulin degludec on liver fat content and abdominal adipose tissue in people with type 2 diabetes (SURPASS-3 MRI): a substudy of the randomised, open-label, parallel-group, phase 3 SURPASS-3 trial. *Lancet Diabetes Endocrinol* 2022;10:393–406 | | 160. Armstrong MJ, Gaunt P, Aithal GP, et al.; LEAN trial team. | Liraglutide safety and efficacy in patients with non-alcoholic steatohepatitis (LEAN): a multicentre, double-blind, randomised, placebo-controlled phase 2 study. *Lancet* 2016;387:679–690 | | 161. Sathyanarayana P, Jogi M, Muthupillai R, Krishnamurthy R, Samson SL, Bajaj M. | Effects of combined exenatide and pioglitazone therapy on hepatic fat content in type 2 diabetes. *Obesity (Silver Spring)* 2011;19:2310–2315 | | 162. Abdul-Ghani M, Migahid O, Megahed A, DeFronzo RA, Al-Ozairi E, Jayyousi A. | Combination therapy with pioglitazone/exenatide improves beta-cell function and produces superior glycaemic control compared with basal/bolus insulin in poorly controlled type 2 diabetes: a 3-year follow-up of the Qatar study. *Diabetes Obes Metab* 2020;22:2287–2294 | | 163. Lavynenko O, Abdul-Ghani M, Alatrach M, et al. | Combination therapy with pioglitazone/exenatide/metformin reduces the prevalence of hepatic fibrosis and steatosis: the efficacy and durability of initial combination therapy for type 2 diabetes (EDICT). *Diabetes Obes Metab* 2022;24:899–907 | | 164. Garvey WT, Frias JP, Jastreboff AM, et al.; SURMOUNT-2 investigators. | Tirzepatide once weekly for the treatment of obesity in people with type 2 diabetes (SURMOUNT-2): a double-blind, randomised, multicentre, placebo-controlled, phase 3 trial. *Lancet* 2023;402:613–626 | | 165. Davies M, Færch L, Jeppesen OK, et al.; STEP 2 Study Group. | Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 2021;397:971–984 | | 166. Bonora E, Frias JP, Tinahones FJ, et al. | Effect of dulaglutide 3.0 and 4.5 mg on weight in patients with type 2 diabetes: exploratory analyses of AWARD-11. *Diabetes Obes Metab* 2021;23:2242–2250 | | 167. Davies MJ, Bergenstal R, Bode B, et al.; NN8022-1922 Study Group. | Efficacy of liraglutide for weight loss among patients with type 2 diabetes: the SCALE diabetes randomized clinical trial. *JAMA* 2015;314:687–699 |

2026 January 29 on guest by

http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S183/848878/dc26s009.pdf from Downloaded

diabetesjournals.org/care # Pharmacologic Approaches to Glycemic Treatment

S213

self-care in people with type 2 diabetes not taking insulin. *Diabetes Technol Ther* 2025;27:261–270 systematic review and meta-analysis of interventional the National Health Interview Survey. *Med Care* 2016;54:796–803

| 196. | Battelino T, Lalic N, Hussain S, et al. The use of continuous glucose monitoring in people living with obesity, intermediate hyperglycemia or type 2 diabetes. *Diabetes Res Clin Pract* 2025;223:112111 || ---- | -----

----- || 197. |
Aronson R, Brown RE, Chu L, et al. Impact of flash glucose monitoring in people with type 2 diabetes inadequately controlled with non-insulin antihyperglycaemic therapy (IMMEDIATE): a randomized controlled trial. *Diabetes Obes Metab* 2023;25:1024–1031 || 198. | Moon SJ, Kim K-S, Lee WJ, Lee MY, Vigersky R, Park C-Y. Efficacy of intermittent short-term use of a real-time continuous glucose monitoring system in non-insulin-treated patients with type 2 diabetes: a randomized controlled trial. *Diabetes Obes Metab* 2023;25:110–120 || 199. | Martens T, Beck RW, Bailey R, et al.; MOBILE Study Group. Effect of continuous glucose monitoring on glycemic control in patients with type 2 diabetes treated with basal insulin: a randomized clinical trial. *JAMA* 2021;325:2262–2272 || 200. | Agarwal S, Varghese RT, Savian R, Low Wang CC, Galindo RJ. Moving beyond glycated hemoglobin to glucose patterns: newer indications for continuous glucose monitor use. *Endocr Pract* 2025;31:1046–1053 || 201. | van Dijk P, Bouma A, Landman GW, et al. Hypoglycemia in frail elderly patients with type 2 diabetes mellitus treated with sulfonylurea. *J Diabetes Sci Technol* 2017;11:438–439 || 202. | Tamborlane WV, Beck RW, Bode BW, et al.; Juvenile Diabetes Research Foundation Continuous Glucose Monitoring Study Group. Continuous glucose monitoring and intensive treatment of type 1 diabetes. *N Engl J Med* 2008;359:1464–1476 || 203. | Hansen KW, Bibby BM. The frequency of intermittently scanned glucose and diurnal variation of glycemic metrics. *J Diabetes Sci Technol* 2022;16:1461–1465 || 204. | Urakami T, Yoshida K, Kuwabara R, et al. Frequent scanning using flash glucose monitoring contributes to better glycemic control in children and adolescents with type 1 diabetes. *J Diabetes Investig* 2022;13:185–190 || 205. | Laffel LM, Kanapka LG, Beck RW, et al.; CGM Intervention in Teens and Young Adults with T1D (CITY) Study Group. Effect of continuous glucose monitoring on glycemic control in adolescents and young adults with type 1 diabetes: a randomized clinical trial. *JAMA* 2020;323:2388–2396 || 206. | New JP, Ajjan R, Pfeiffer AFH, Freckmann G. Continuous glucose monitoring in people with diabetes: the randomized controlled Glucose Level Awareness in Diabetes Study (GLADIS). *Diabet Med* 2015;32:609–617 || 207. | Gubitosi-Klug RA, Braffett BH, Bebu I, et al. Continuous glucose monitoring in adults with type 1 diabetes with 35 years duration from the DCCT/EDIC study. *Diabetes Care* 2022;45:659–665 || 208. | Teo E, Hassan N, Tam W, Koh S. Effectiveness of continuous glucose monitoring in maintaining glycaemic control among people with type 1 diabetes mellitus: a systematic review of randomised controlled trials and meta-analysis. *Diabetologia* 2022;65:604–619 || 209. | Seidu S, Kunutsor SK, Ajjan P. Efficacy and safety of continuous glucose monitoring and intermittently scanned continuous glucose monitoring in patients with type 2 diabetes: a systematic review and meta-analysis. *Diabetes Care* 2025;48:177–181 |

2026 January 29 on guest by

http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S183/848878/dc26s009.pdf from Downloaded

Pharmacologic Approaches to Glycemic Treatment

Diabetes Care Volume 49, Supplement 1, January 2026

1. National Library of Medicine. National Center for Biotechnology Information. ClinicalTrials.gov. A Study of the Effect of Tirzepatide on How the Body Handles Birth Control Pills in Healthy Female Participants (NCT04172987). Accessed 12 June 2024. Available from <https://clinicaltrials.gov/study/NCT04172987> 2. U.S. Food & Drug Administration. Tirzepatide prescribing information. Accessed 25 August 2025. Available from https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/215866s000lbl.pdf 3. Chang L-S, Barroso-Sousa R, Tolaney SM, Hodi FS, Kaiser UB, Min L. Endocrine toxicity of cancer immunotherapy targeting immune checkpoints. *Endocr Rev* 2019;40:17–65 4. Zhou L, Yang S, Li Y, Xue C, Wan R. A comprehensive review of immune checkpoint inhibitor-related diabetes mellitus: incidence, clinical features, management, and prognosis. *Front Immunol* 2024;15:1448728 5. Barroso-Sousa R, Barry WT, Garrido-Castro AC, et al. Incidence of endocrine dysfunction following the use of different immune checkpoint inhibitor regimens: a systematic review and meta-analysis. *JAMA Oncol* 2018;4:173–182 6. Liu J, Zhou H, Zhang Y, et al. Reporting of immune checkpoint inhibitor therapy-associated diabetes, 2015–2019. *Diabetes Care* 2020;43:e79–e80 7. Mulla K, Farag S, Moore B, et al. Hyperglycaemia following immune checkpoint inhibitor

therapy-Incidence, aetiology and assessment. *Diabet Med* 2023;40:e15053 8. Byun DJ, Braunstein R, Flynn J, et al. Immune checkpoint inhibitor-associated diabetes: a single-institution experience. *Diabetes Care* 2020;43:3106–3109 9. Schneider BJ, Naidoo J, Santomaso BD, et al. Management of immune-related adverse events in patients treated with immune checkpoint inhibitor therapy: ASCO guideline update. *J Clin Oncol* 2021;39:4073–4126 10. Goncalves MD, Hopkins BD, Cantley LC. phosphatidylinositol 3-kinase, growth disorders, and cancer. *N Engl J Med* 2018;379:2052–2062 11. André F, Ciruelos E, Rubovszky G, et al.; SOLAR-1 Study Group. Alpelisib for PIK3CA-mutated, hormone receptor-positive advanced breast cancer. *N Engl J Med* 2019;380:1929–1940 12. André F, Ciruelos EM, Juric D, et al. Alpelisib plus fulvestrant for PIK3CA-mutated, hormone receptor-positive, human epidermal growth factor receptor-2-negative advanced breast cancer: final overall survival results from SOLAR-1. *Ann Oncol* 2021;32:208–217 13. Rugo HS, André F, Yamashita T, et al. Time course and management of key adverse events during the randomized phase III SOLAR-1 study of PI3K inhibitor alpelisib plus fulvestrant in patients with HR-positive advanced breast cancer. *Ann Oncol* 2020;31:1001–1010 14. Jhaveri KL, Im S-A, Saura C, et al. Overall survival with inavolisib in PIK3CA-mutated advanced breast cancer. *N Engl J Med* 2025;393:151–161 15. Liu D, Weintraub MA, Garcia C, et al. Characterization, management, and risk factors of hyperglycemia during PI3K or AKT inhibitor treatment. *Cancer Med* 2022;11:1796–1804 16. Gallagher EJ, Moore H, Lacouture ME, et al. Managing hyperglycemia and rash associated with alpelisib: expert consensus recommendations using the Delphi technique. *NPJ Breast Cancer* 2024;10:12 17. Turk T, Pietruck F, Dolff S, et al. Repaglinide in the management of new-onset diabetes mellitus after renal transplantation. *Am J Transplant* 2006;6:842–846 18. Ode KL, Ballman M, Battezzati A, et al. ISPAD clinical practice consensus guidelines 2022: management of cystic fibrosis-related diabetes in children and adolescents. *Pediatr Diabetes* 2022;23:1212–1228 19. Kurnikowski A, Nordheim E, Schwaiger E, et al. Criteria for prediabetes and posttransplant diabetes mellitus after kidney transplantation: a 2-year diagnostic accuracy study of participants from a randomized controlled trial. *Am J Transplant* 2022;22:2880–2891 20. Sharif A, Chakker A, de Vries APJ, et al. International consensus on post-transplantation diabetes mellitus. *Nephrol Dial Transplant* 2024;39:531–549 21. Kuningas K, Driscoll J, Mair R, et al. Comparing glycaemic benefits of active versus passive lifestyle intervention in kidney allograft recipients: a randomized controlled trial. *Transplantation* 2020;104:1491–1499 22. Hecking M, Haidinger M, Doller D, et al. Early basal insulin therapy decreases new-onset diabetes after renal transplantation. *J Am Soc Nephrol* 2012;23:739–749 23. Schwaiger E, Krenn S, Kurnikowski A, et al. Early postoperative basal insulin therapy versus standard of care for the prevention of diabetes mellitus after kidney transplantation: a multicenter randomized trial. *J Am Soc Nephrol* 2021;32:2083–2098 24. Chandra A, Rao N, Pooniya V, Singh A. Hypoglycemia with insulin in post-transplant diabetes mellitus. *Transpl Immunol* 2023;78:101833 25. Tuerk TR, Bandur S, Nuernberger J, et al. Gliquidone therapy of new-onset diabetes mellitus after kidney transplantation. *Clin Nephrol* 2008;70:26–32 26. Lo C, Toyama T, Oshima M, et al. Glucose-lowering agents for treating pre-existing and new-onset diabetes in kidney transplant recipients. *Cochrane Database Syst Rev* 2020;8:CD009966 27. Lawrence SE, Chandran MM, Park JM, et al. Sweet and simple as syrup: a review and guidance for use of novel antihyperglycemic agents for

2026 January 29 on guest by

http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S183/848878/dc26s009.pdf from Downloaded

diabetesjournals.org/care # Pharmacologic Approaches to Glycemic Treatment

S215

post-transplant diabetes mellitus and type 2 transplant recipients with diabetes. *Clin Transplant* 309. Sánchez Fructuoso AI, Bedia Raba A, Banegas diabetes mellitus after kidney transplantation. 2025;39:e70144 Deras E, et al. Sodium-glucose cotransporter-2 *Clin Transplant* 2023;37:e14922

287. Munoz Pena JM, Cusi K. Posttransplant diabetes mellitus: recent developments in pharmacological management of hyperglycemia. *J Clin Endocrinol Metab* 2023;109:e1–e11

288. Ram E, Lavee J, Tenenbaum A, et al. Metformin therapy in patients with diabetes mellitus is associated with a reduced risk of vasculopathy and cardiovascular mortality after heart transplantation. *Cardiovasc Diabetol* 2019;18:118

289. Vest LS, Korashy FM, Zhang Z, et al. Metformin use in the first year after kidney transplant, correlates, and associated outcomes in diabetic transplant recipients: a retrospective analysis of integrated registry and

pharmacy claims data. Clin Transplant 2018;32:e13302

290. Abdelaziz TS, Ali AY, Fathy M. Efficacy and safety of dipeptidyl peptidase-4 inhibitors in kidney transplant recipients with post-transplant diabetes mellitus (PTDM)- a systematic review and meta-analysis. Curr Diabetes Rev 2020;16:580–585

291. Thangavelu T, Lyden E, Shivaswamy V. A retrospective study of glucagon-like peptide 1 receptor agonists for the management of diabetes after transplantation. Diabetes Ther 2020;11:987–994

292. Kukla A, Hill J, Merzkani M, et al. The use of GLP1R agonists for the treatment of type 2 diabetes in kidney transplant recipients. Transplant Direct 2020;6:e524

293. Singh P, Pesavento TE, Washburn K, Walsh D, Meng S. Largest single-centre experience of dulaglutide for management of diabetes mellitus in solid organ transplant recipients. Diabetes Obes Metab 2019;21:1061–1065

294. Liou J-H, Liu Y-M, Chen C-H. Management of diabetes mellitus with glucagonlike peptide-1 agonist liraglutide in renal transplant recipients: a retrospective study. Transplant Proc 2018;50:2502–2505

295. Vigara LA, Villanego F, Orellana C, et al. Effectiveness and safety of glucagon-like peptide-1 receptor agonist in a cohort of kidney transplant recipients. Clin Transplant 2022;36:e14633

296. Sweiss H, Hall R, Zeilmann D, et al. Single-center evaluation of safety & efficacy of glucagon-like peptide-1 receptor agonists in solid organ transplantation. Prog Transplant 2022;32:357–362

297. Zelada H, Campana M, Kawai K, et al. Efficacy, tolerability, and safety of glucagon-like peptide 1 receptor agonists (GLP1-RA) in kidney transplant recipients.

298. Kim HS, Lee J, Jung CH, Park J-Y, Lee WJ. Dulaglutide as an effective replacement for prandial insulin in kidney transplant recipients with type 2 diabetes mellitus: a retrospective review. Diabetes Metab J 2021;45:948–953

299. Orandi BJ, Chen Y, Li Y, et al. GLP-1 receptor agonists in kidney transplant recipients with pre-existing diabetes: a retrospective cohort study. Lancet Diabetes Endocrinol 2025;13:374–383

300. Lin L-C, Chen J-Y, Huang TT-M, Wu V-C. Association of glucagon-like peptide-1 receptor agonists with cardiovascular and kidney outcomes in type 2 diabetic kidney transplant recipients. Cardiovasc Diabetol 2025;24:87

301. Diker Cohen T, Rudman Y, Turjeman A, et al. Glucagon-like peptide 1 receptor agonists and renal outcomes in kidney transplant recipients with diabetes mellitus. Diabetes Metab 2025;51:101624

302. Dotan I, Rudman Y, Turjeman A, et al. Glucagon-like peptide 1 receptor agonists and cardiovascular outcomes in solid organ transplant recipients with diabetes mellitus. Transplantation 2024;108:e121–e128

303. Sammour Y, Nassif M, Magwire M, et al. Effects of GLP-1 receptor agonists and SGLT-2 inhibitors in heart transplant patients with type 2 diabetes: initial report from a cardiometabolic center of excellence. J Heart Lung Transplant 2021;40:426–429

304. Lim J-H, Kwon S, Jeon Y, et al. The efficacy and safety of SGLT2 inhibitor in diabetic kidney transplant recipients. Transplantation 2022;106:e404–e412

305. Schwaiger E, Burghart L, Signorini L, et al. Empagliflozin in posttransplantation diabetes mellitus: a prospective, interventional pilot study on glucose metabolism, fluid volume, and patient safety. Am J Transplant 2019;19:907–919

306. Halden TAS, Kvitne KE, Midtvedt K, et al. Efficacy and safety of empagliflozin in renal transplant recipients with posttransplant diabetes mellitus. Diabetes Care 2019;42:1067–1074

307. Mahling M, Schork A, Nadalin S, Fritsche A, Heyne N, Guthoff M. Sodium-glucose cotransporter 2 (SGLT2) inhibition in kidney transplant recipients with diabetes mellitus. Kidney Blood Press Res 2019;44:984–992

308. Shah M, Virani Z, Rajput P, Shah B. Efficacy and safety of canagliflozin in kidney transplant patients. *Indian J Nephrol* 2019;29:278–281

2026 January 29 on guest by

http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S183/848878/dc26s009.pdf from Downloaded